Optoelectronics GROUP @ ZJU

浙江大学 ZHEJIANG UNIVERSITY



Fundamentals of Optoelectronics (光电子学基础)

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2025 春夏学期

Fourier Transform



In 1821, Joseph Fourier claimed that any function, whether continuous or discontinuous, can be expanded into a series of sines.

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

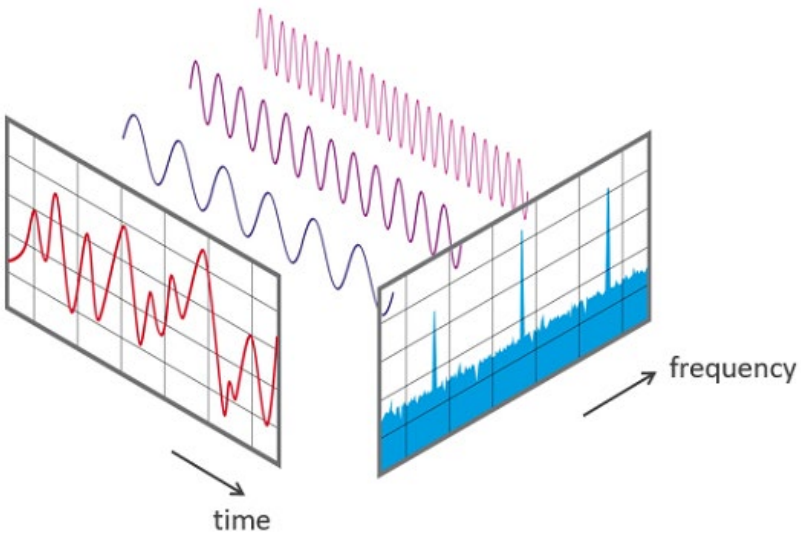
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

"Fourier's theorem is not only one of the most beautiful results of modern analysis, but it may be said to furnish an indispensable instrument in the treatment of nearly every recondite question in modern physics."

Lord Kelvin
(1824 – 1907)



Fourier Transform



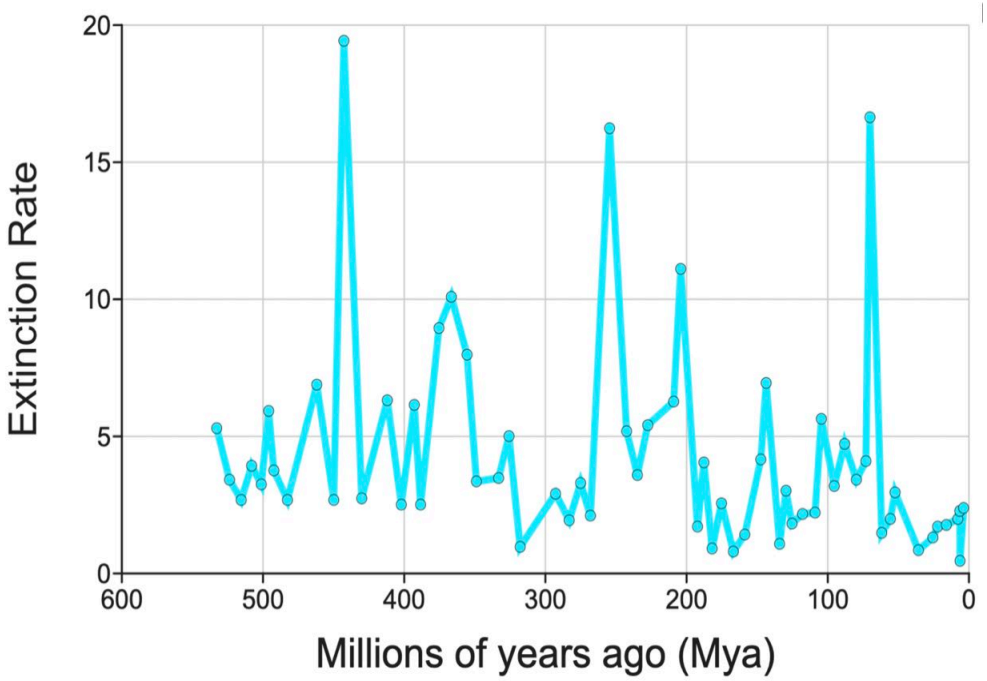
Fourier analysis is one of the most important mathematical techniques, maybe the most important mathematical technique, ever developed for science. Nearly every single field of science benefits from it.

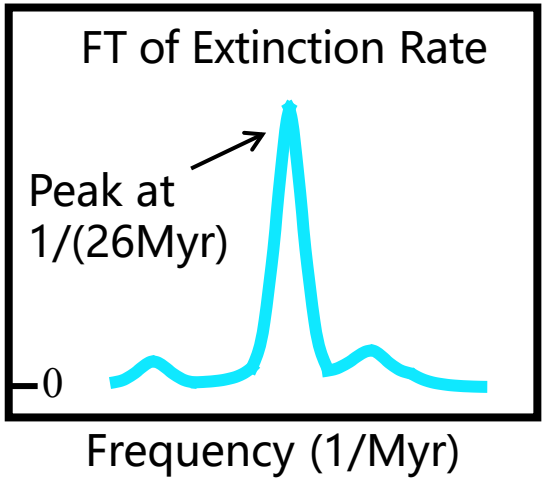
In simplest terms, the Fourier transform is an approach for computing what frequencies are present in a wave—in particular, the spectrum—from its dependence on time.

Some examples of Fourier Transform

Archaeologists know the species extinction rate over many million years.

So, when can we expect the next one?



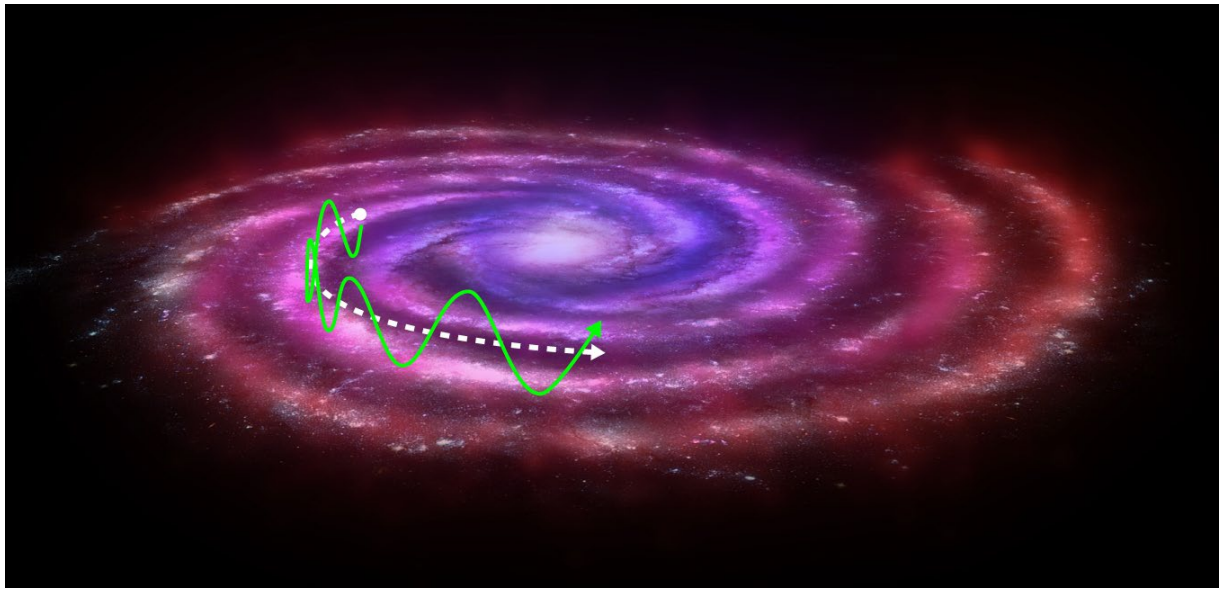


Why 26 million years?

Why would asteroids and comets tend to strike earth every 26 million years?

Our solar system revolves（绕转） around the center of our galaxy.

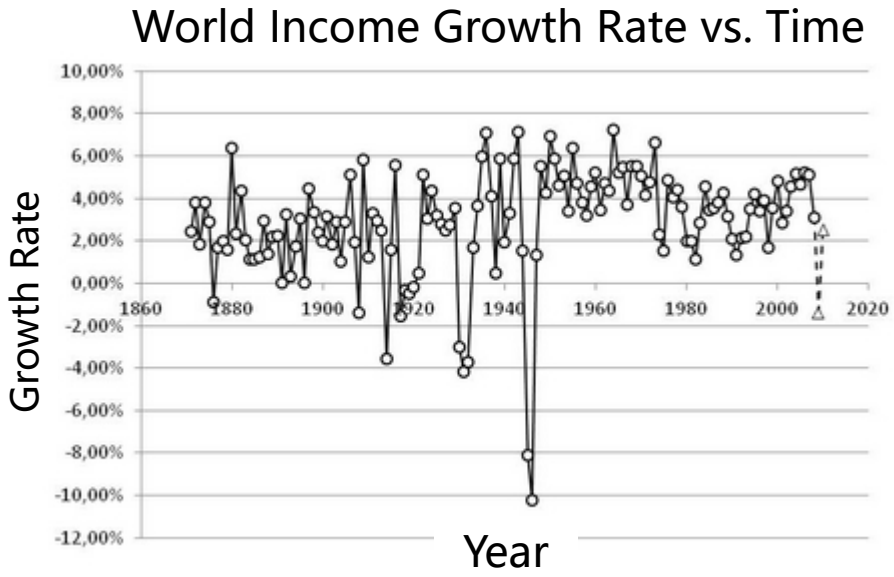
But it also oscillates through the galactic plane（银道面）—with a 52 million-year period.



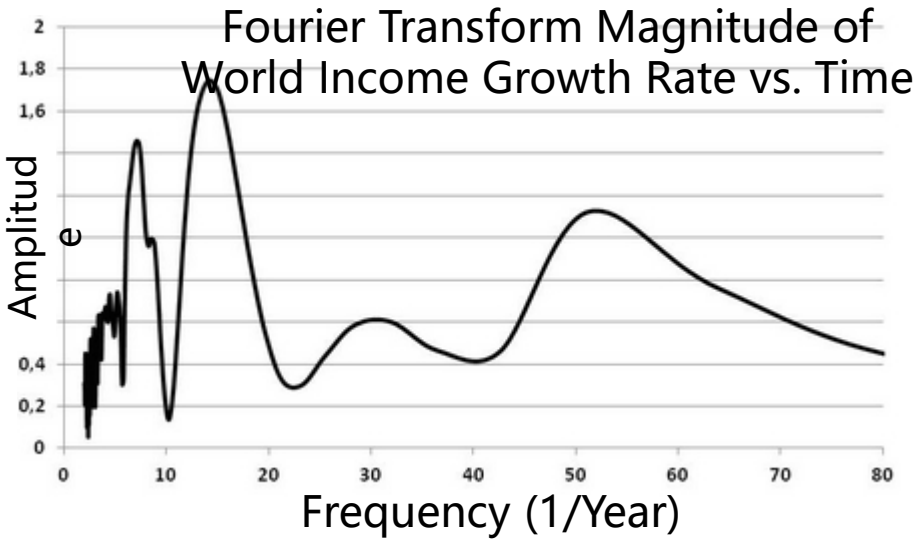
Every 26 million years, the earth crosses the galactic plane—encountering much higher-density regions of the galaxy and hence many more asteroids.

Some examples of Fourier Transform

World’s Income Growth Rate vs. Time and Its Fourier Transform



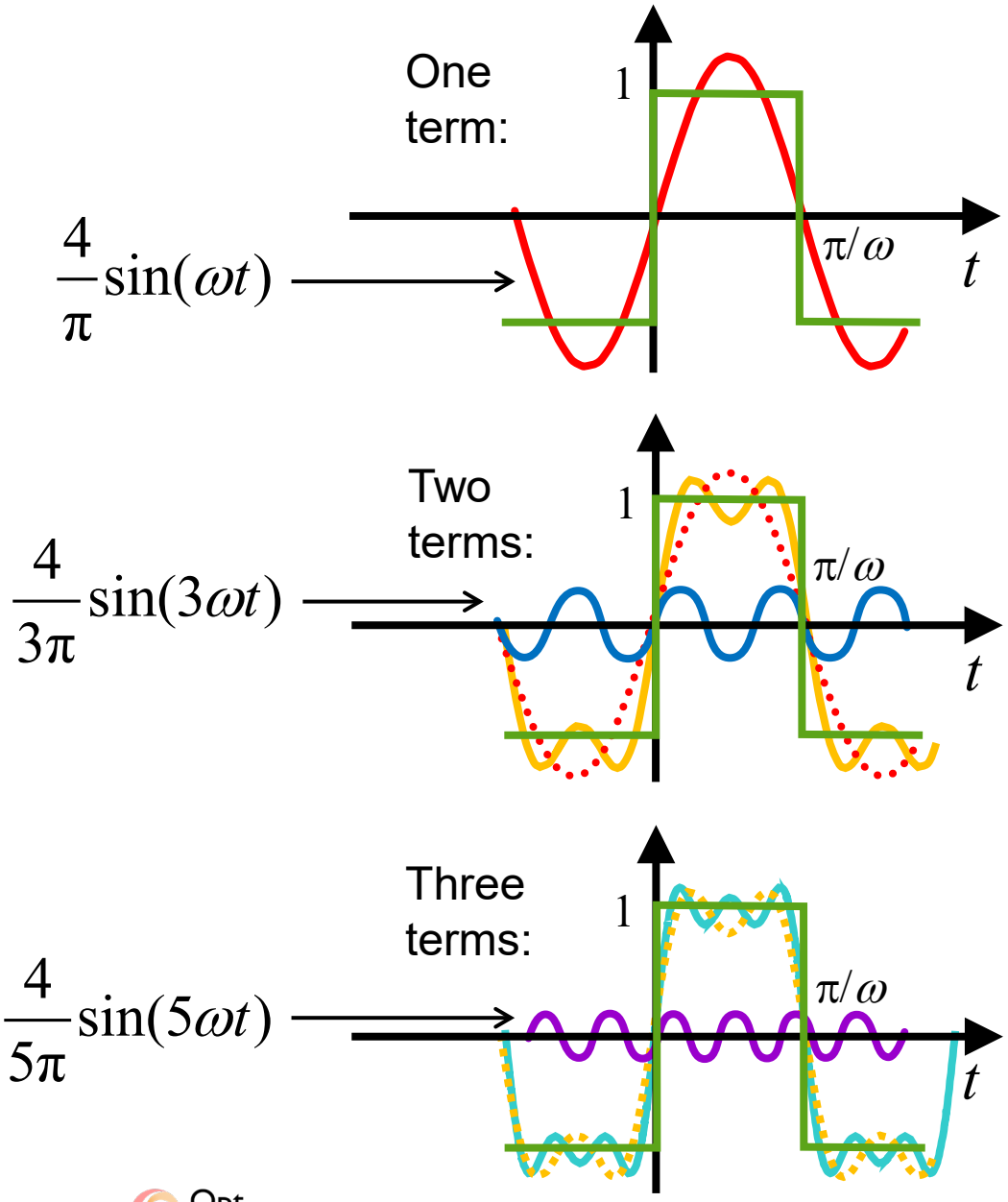
Knowledge of preferred frequencies could help governments prevent crashes and individuals invest wisely.



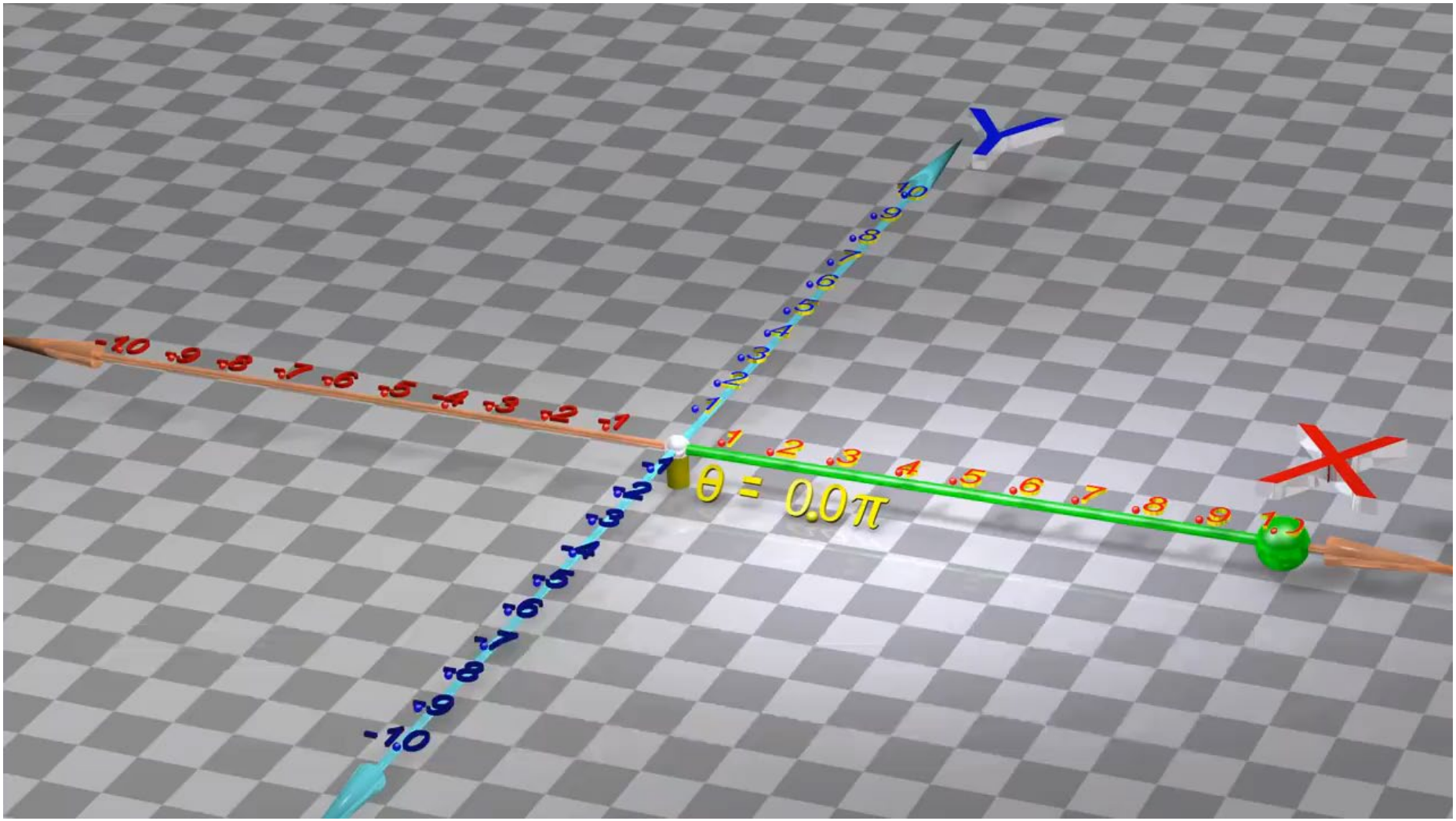
Fourier Decomposing Functions

Here, we write a **square wave** as a sum of sine waves of different frequencies.

The sharp corners require very high frequencies.



Fourier series

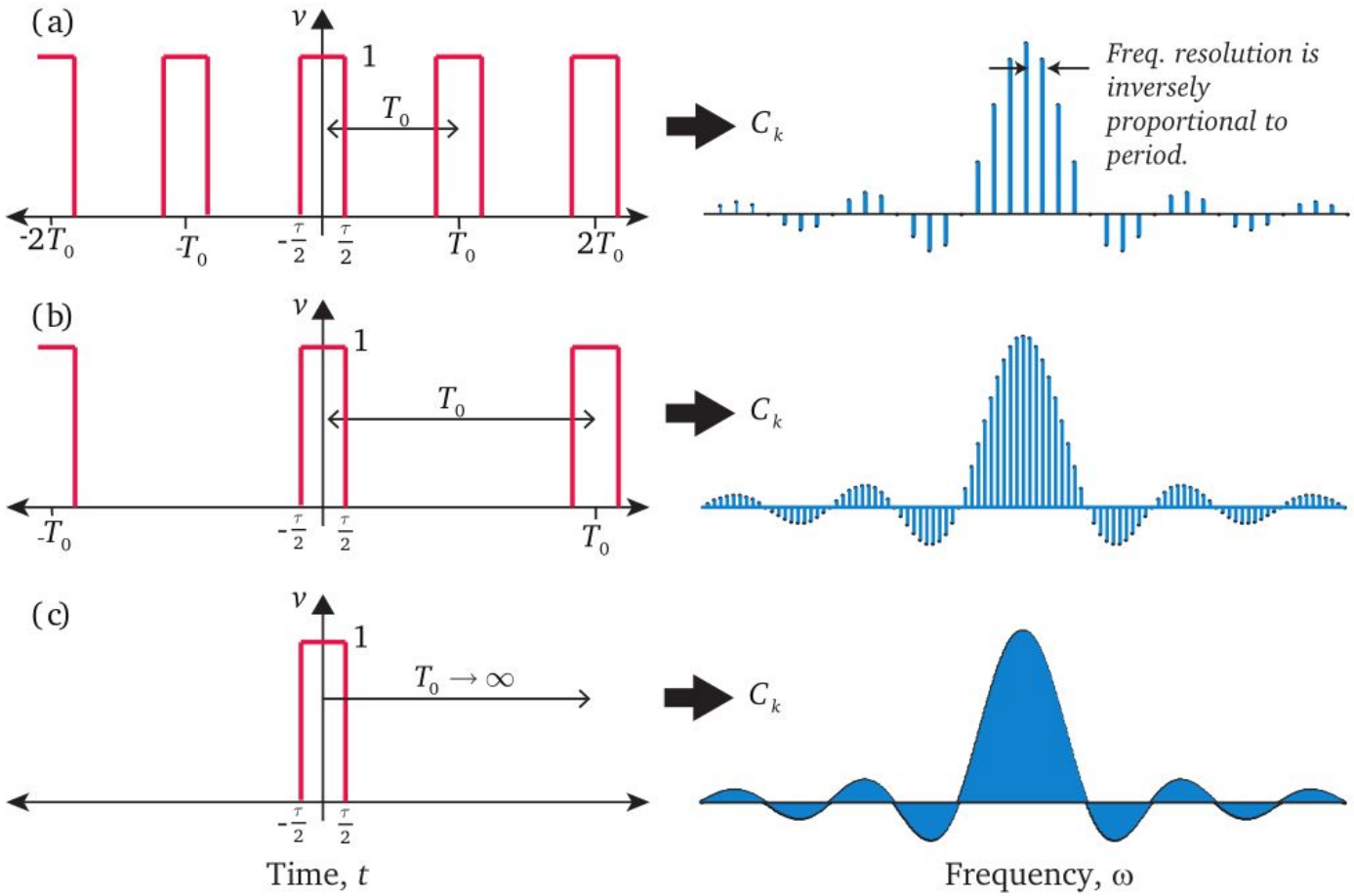


e指数是旋转，三角函数是波，两个同频的三角函数合成出来还是三角函数，解释光在玻璃里变慢原因

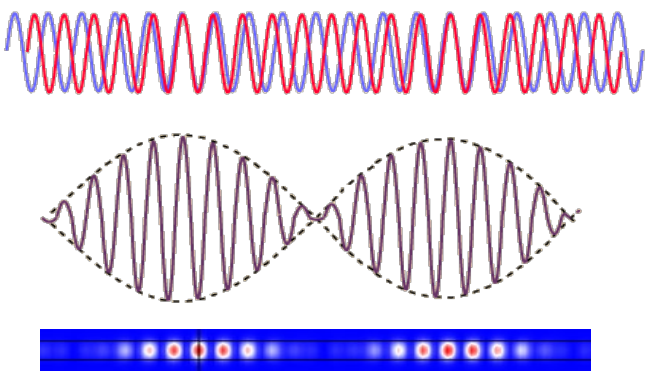
到10:17秒停止

Fourier series and Fourier transform

Fourier series for periodic functions, and Fourier transform for functions that are not periodic. The Fourier transform is a generalization of Fourier series.



Essence of Fourier Transform



拍频（Beat frequency）现象是指当两个频率接近但不完全相同的波形相互叠加时，产生的结果波形会显示出一种振幅随时间缓慢变化的模式。这种效果源于两个波形的相位差随时间变化，导致在某些时刻它们相长干涉（增强），而在其他时刻相消干涉（减弱）。

当两个频率相近的信号混合时，它们的拍频可以用来提取原始信号的频率和相位信息。

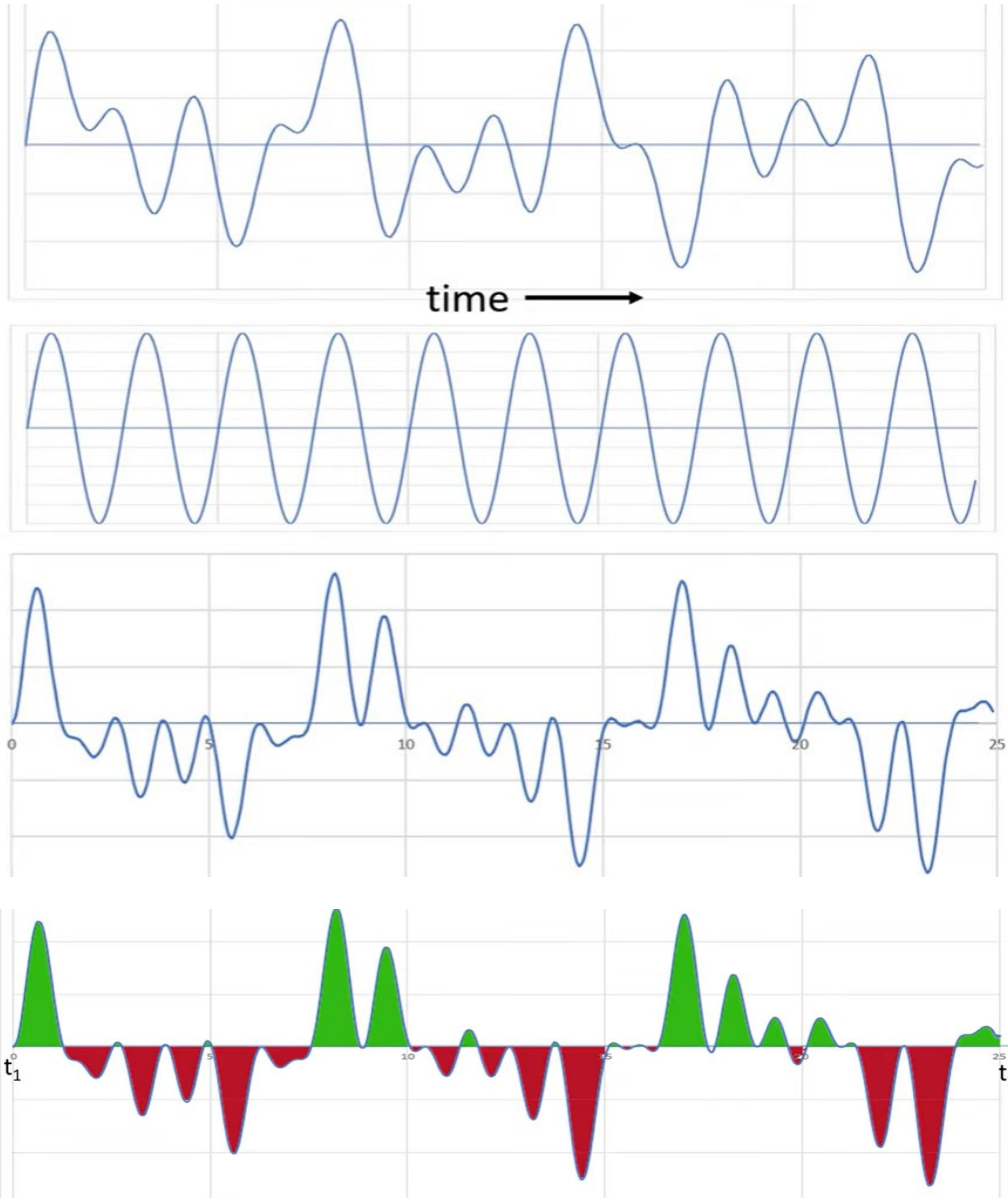
Essence of Fourier Transform

Signal

X

Frequency f_1

=



Integrate all the values over time.

If the frequency under examination is not present in the signal, the sum will be zero because the positive and negative values will cancel each other out over time.

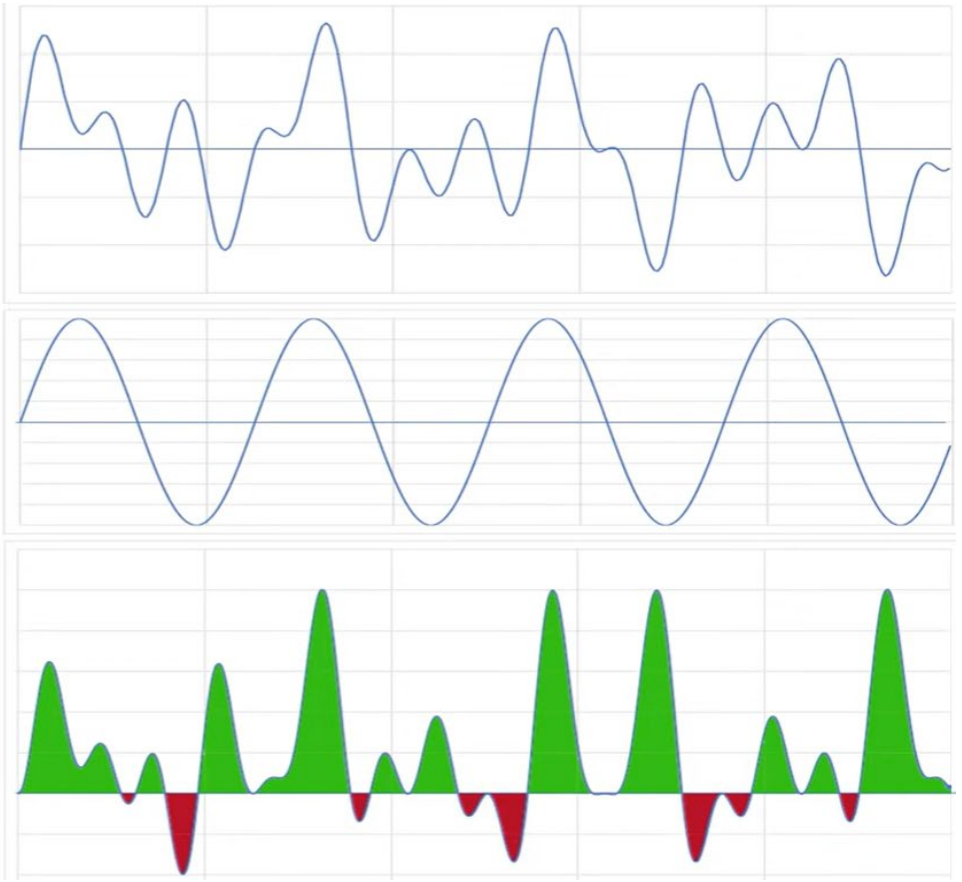
Essence of Fourier Transform

Same signal

X

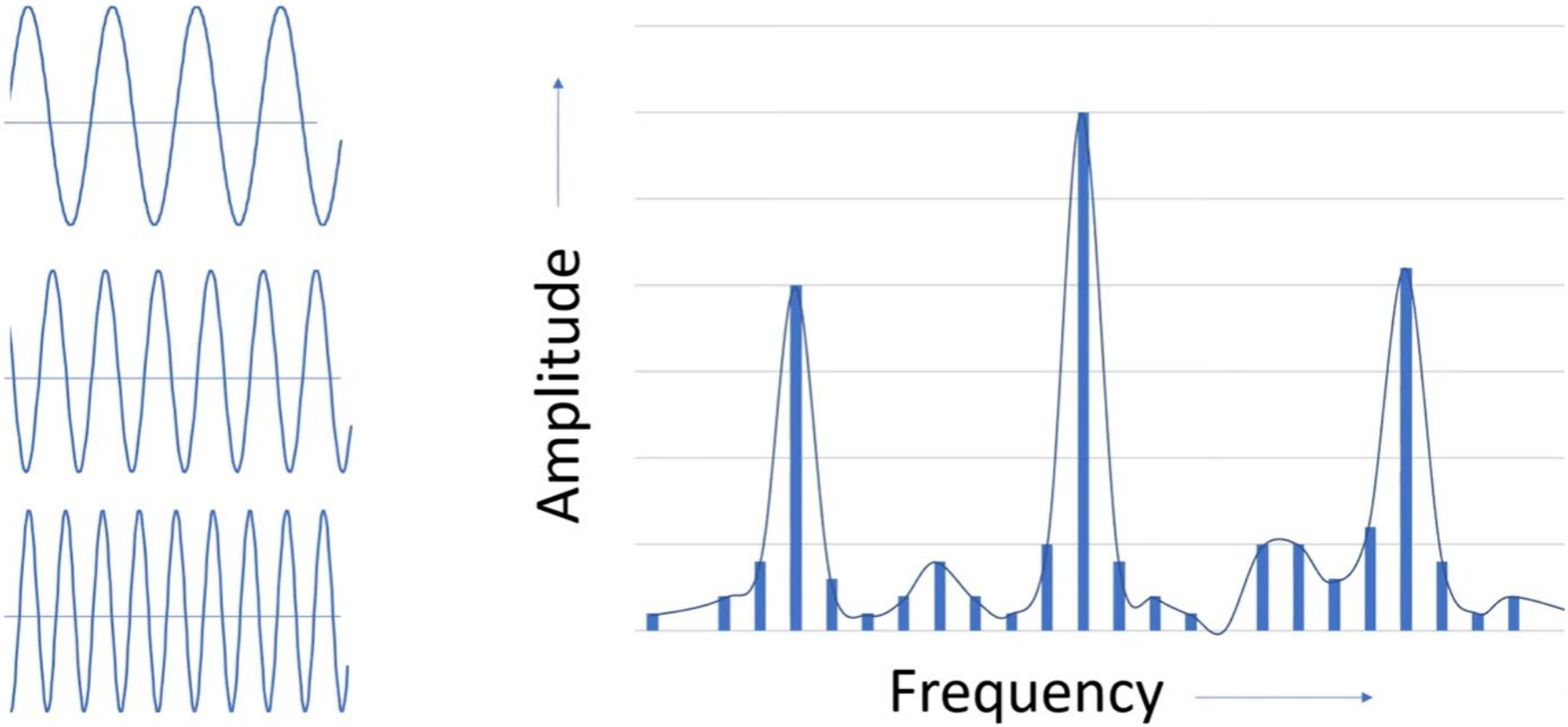
Frequency f_2

=



Do the same integration for a frequency which actually is present in the signal, the net value will be much larger.

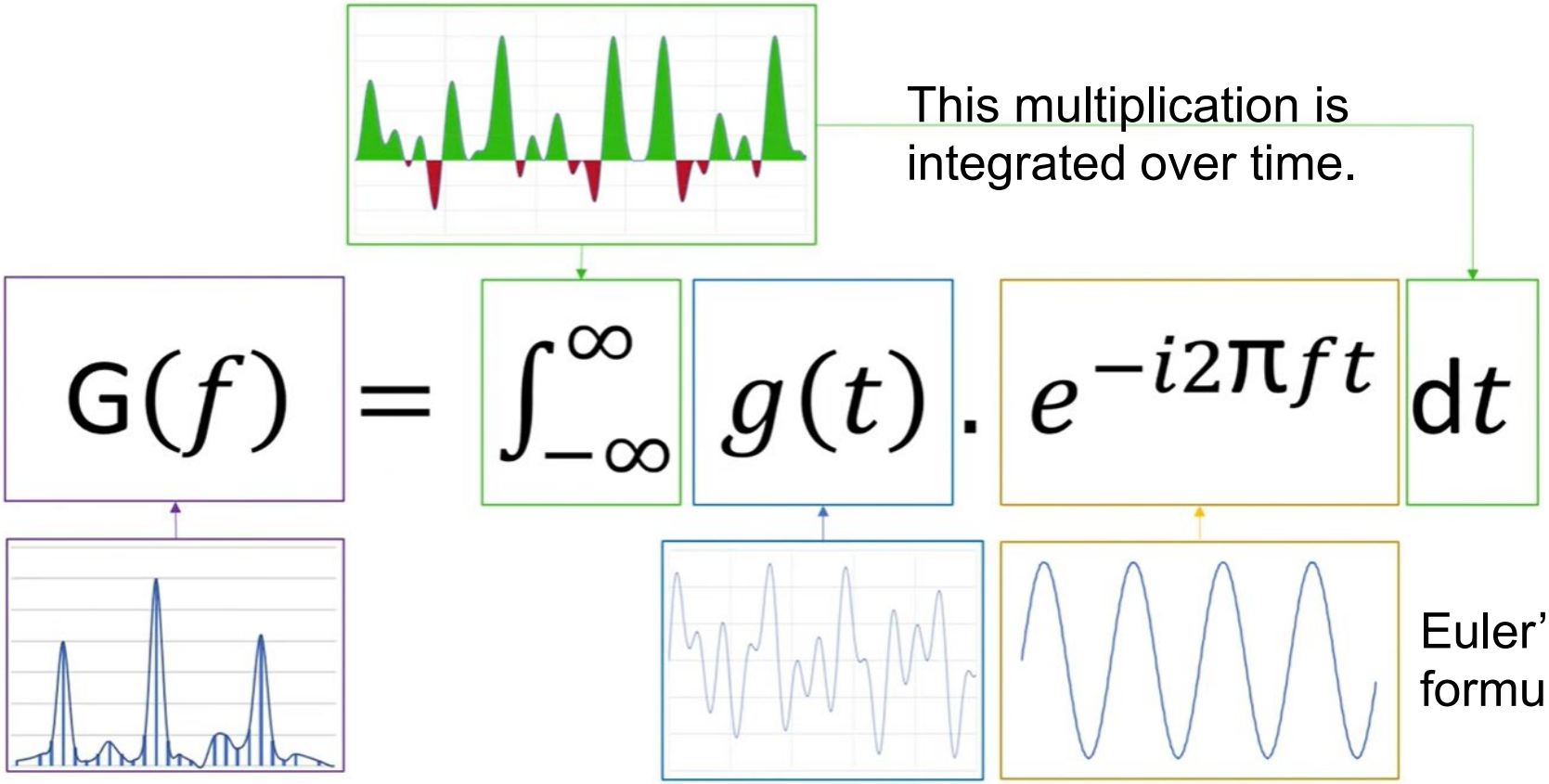
Essence of Fourier Transform



By repeating this multiplication and integration process for a large number of frequencies, we can construct a frequency spectrum.

Essence of Fourier Transform

This process can be described by the Fourier Transform formula.

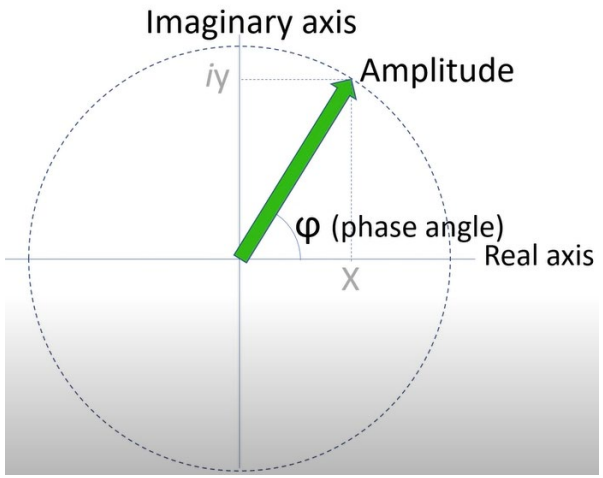


Spectrum as a function of the frequency **G(*f*)**

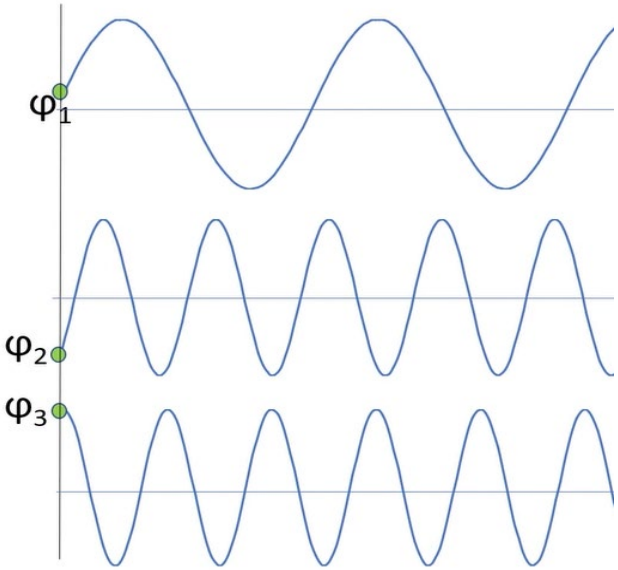
Signal as a function of time **g(*t*),** which includes phase φ of each frequency

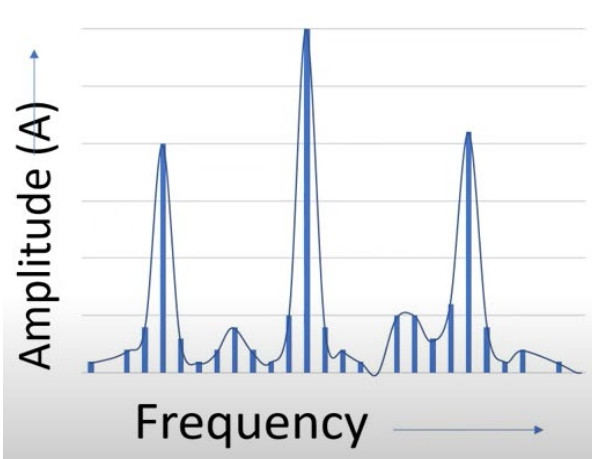
Complex harmonic function with frequency ***f*** , also include phase information

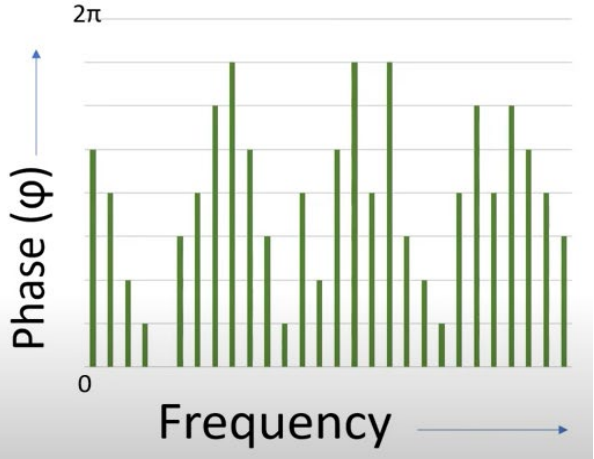
Essence of Fourier Transform



$$e^{i\varphi} = cos\varphi + isin\varphi$$

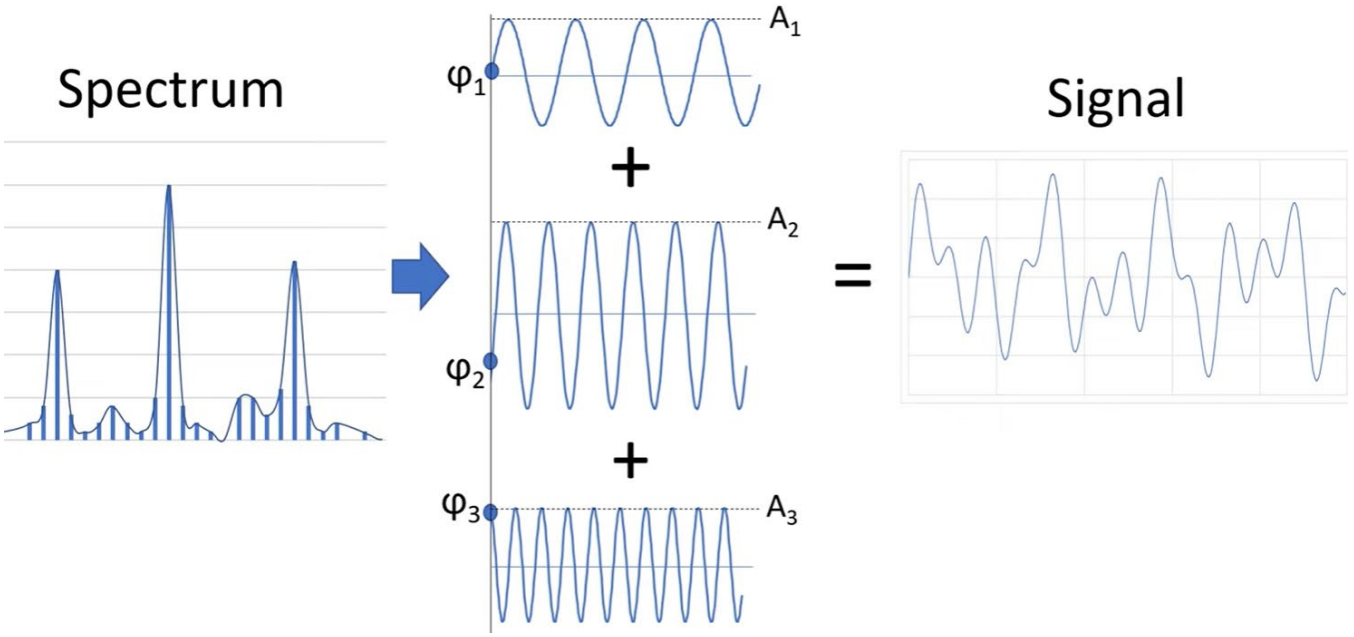






Fourier transform contains both amplitude and phase information.

Inverse Fourier Transform



$$F(\omega) = \int_{-\infty}^{\infty} f(t) \exp(-i\omega t)\, dt$$

Fourier Transform

So, we can transform to the frequency domain and back. Interestingly, these transformations are very similar.

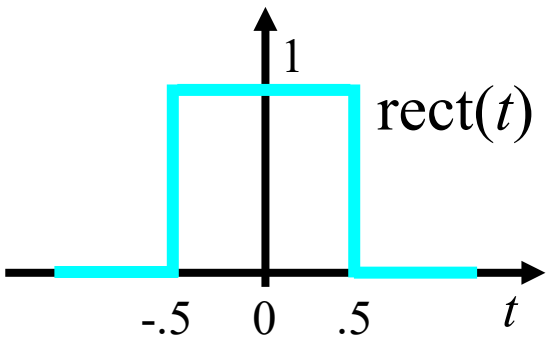
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \exp(i\omega t)\, d\omega$$

Inverse Fourier Transform

These equations together are Fourier's Theorem.

Fourier Transform of a Rectangle Function, rect(t)

$$F(\omega) = \int_{-\infty}^{\infty} \boxed{f(t)}^{=1} \exp(-i\omega t)\, dt$$



$$F(\omega) = \int_{-1/2}^{1/2} \exp(-i\omega t)\, dt = \frac{1}{-i\omega}[\exp(-i\omega t)]_{-1/2}^{+1/2}$$

$$= \frac{1}{-i\omega}[\exp(-i\omega/2) - \exp(i\omega/2)]$$

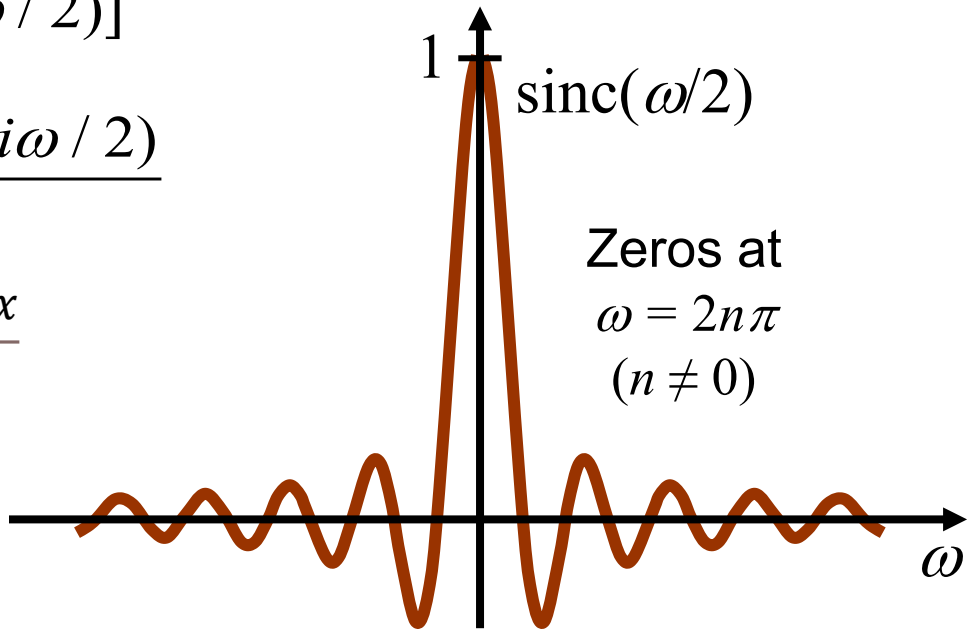
$$= \frac{1}{(\omega/2)} \frac{\exp(i\omega/2) - \exp(-i\omega/2)}{2i}$$

$$= \frac{\sin(\omega/2)}{(\omega/2)} \qquad \operatorname{sinc} x = \frac{\sin x}{x}$$

$$\equiv \operatorname{sinc}(\omega/2)$$

$$\boxed{F(\omega) = \sin\ c(\omega/2)}$$

The sinc function is important in optics.

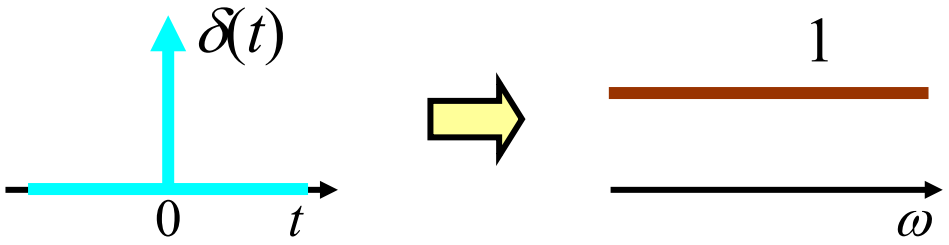


Note that *F* extends out to ±∞ due to the discontinuities in rect(*t*).

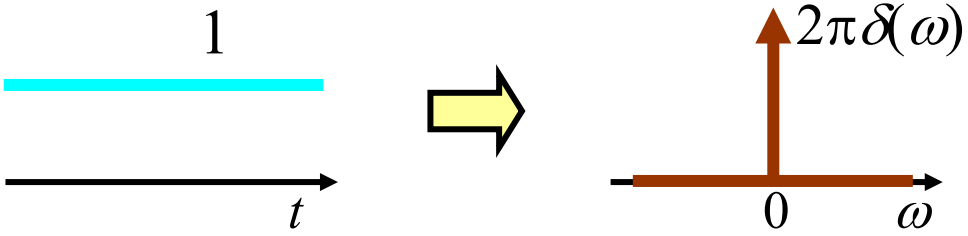
The Fourier transform of Dirac δ-Function $\delta(t)$ is 1

$$\int_{-\infty}^{\infty} \delta(t)\exp(-i\omega t)\,dt \;=\; \exp[-i\omega(0)] \;=\; 1$$

$$\boxed{\int_{-\infty}^{\infty} \delta(t)\,dt = 1}$$



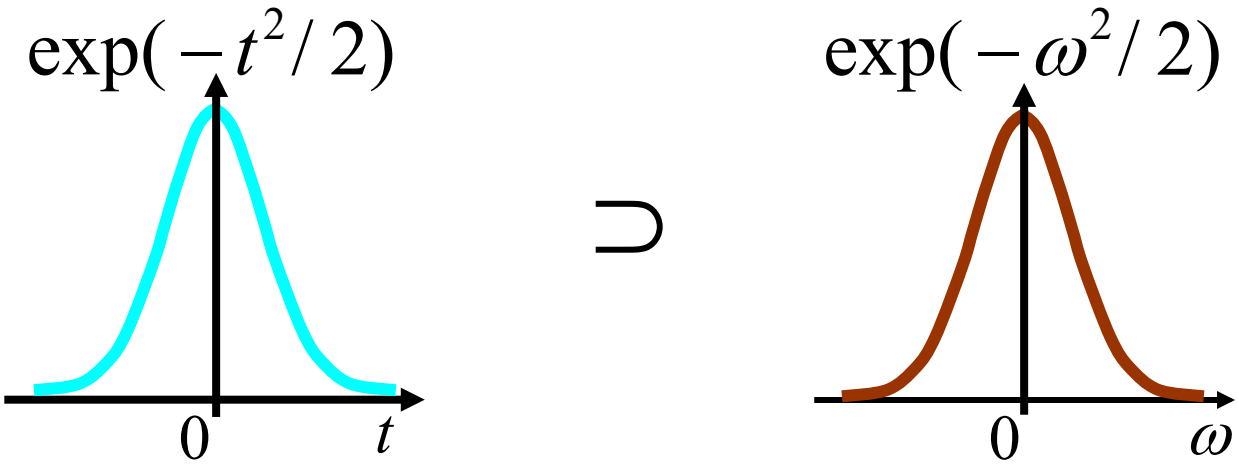
And the Fourier Transform of 1 is $2\pi\delta(\omega)$: $\int_{-\infty}^{\infty} 1\exp(-i\omega t)\,dt \;=\; 2\pi\,\delta(\omega)$



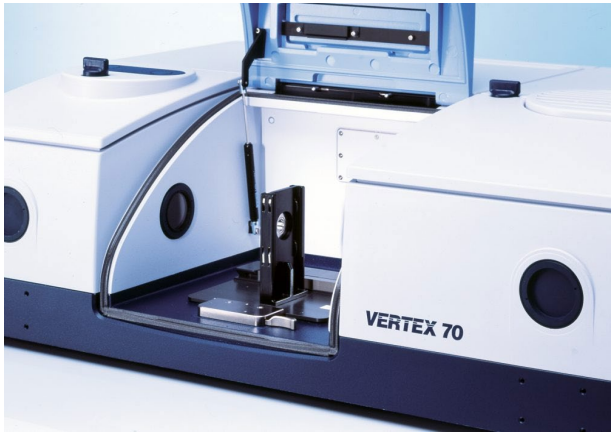
Fourier transform of a Gaussian, exp(-t²/2), is itself!

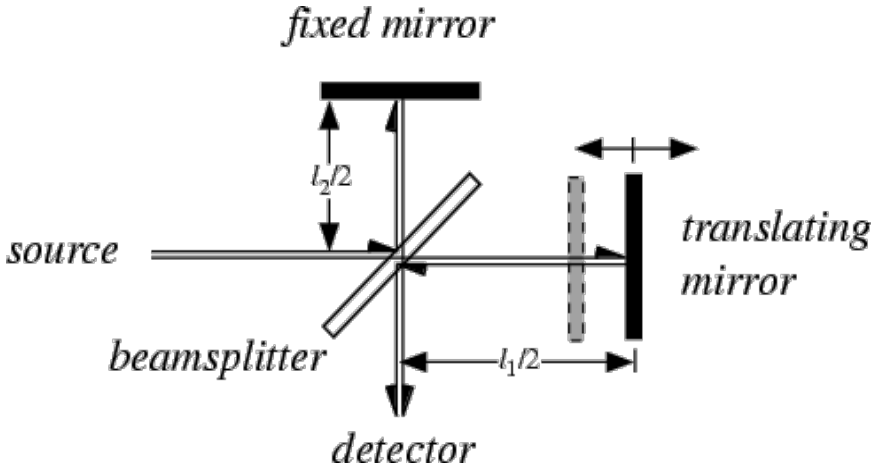
$$\mathrm{F}\quad \{\exp(-t^2/2)\} = \int_{-\infty}^{\infty} \exp(-t^2/2)\exp(-i\omega t)\,dt$$

正比与 $\propto \exp(-\omega^2/2)$



Example：Fourier Transform Spectroscopy

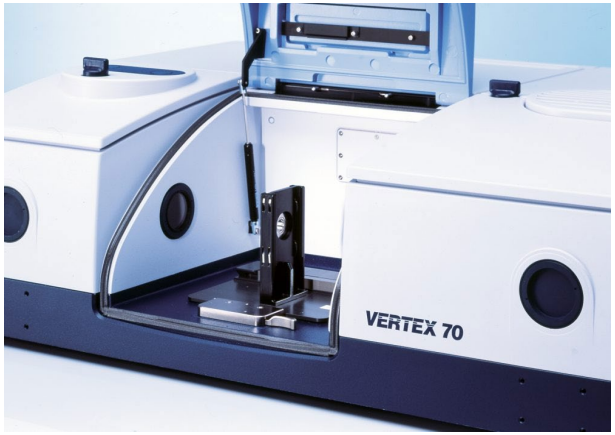


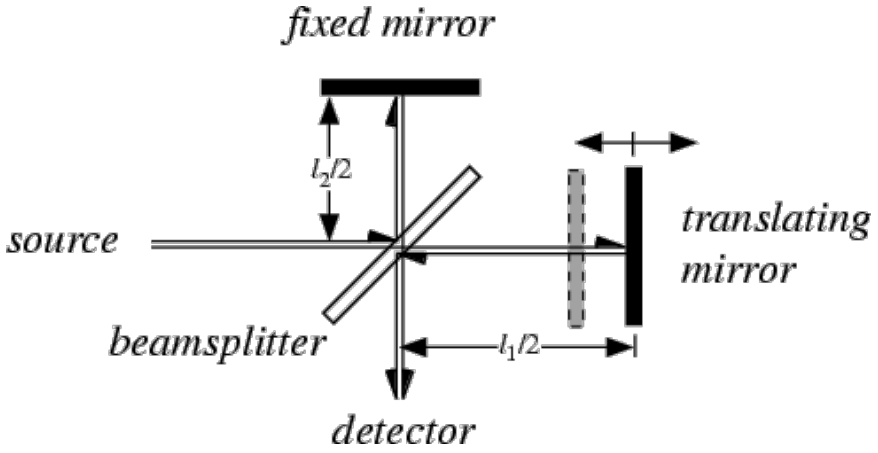


对于单色光来说，若其入射波长为λ，波数为σ（σ=1/λ），入射强度为2B(σ)，经过半透半反镜后，光束分为强度相等（B(σ)）的两束，假设两路光程差为x，当x改变时，其干涉强度关于波数呈余弦函数关系，则像面上两点的强度可以表示为：

$$I(x) = 2B(\sigma)[1 + \cos(2\pi x\sigma)]$$ 干涉公式

Example：Fourier Transform Spectroscopy



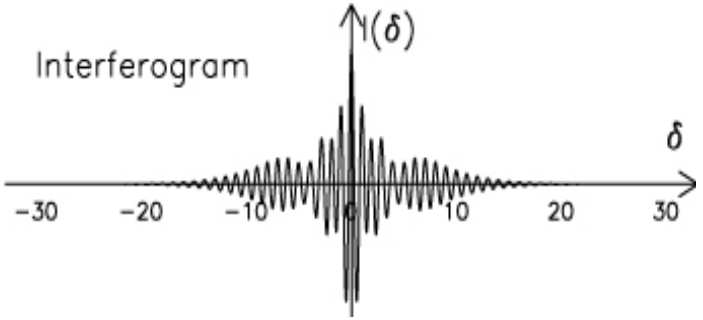


当光源中含有多种频率的光时，探测器上光强可表示为

$$I(x) = \int_0^\infty 2B(\sigma)[1 + \cos(2\pi x\sigma)]d\sigma$$

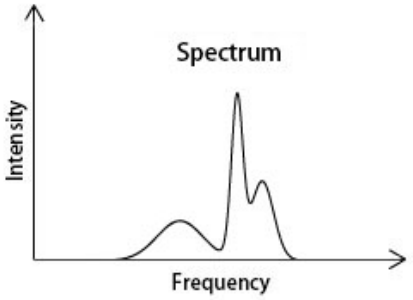
减去强度的平均值(偏置)$2B(\sigma)$，得到

$$I(x) = \int_0^\infty 2B(\sigma)\cos(2\pi x\sigma)\,d\sigma$$

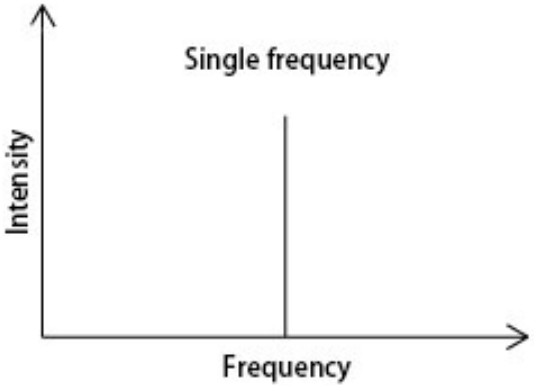


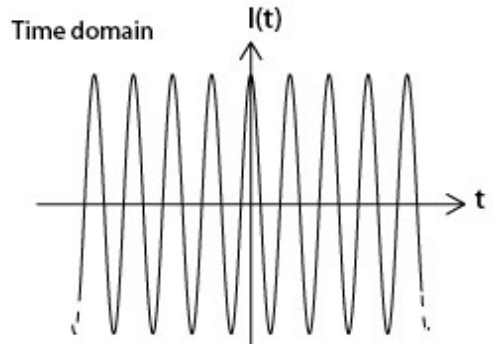
利用傅里叶反变换在计算机上求解得到光谱，由于B(σ)系数不影响光谱成分，可去除2，反变换公式：

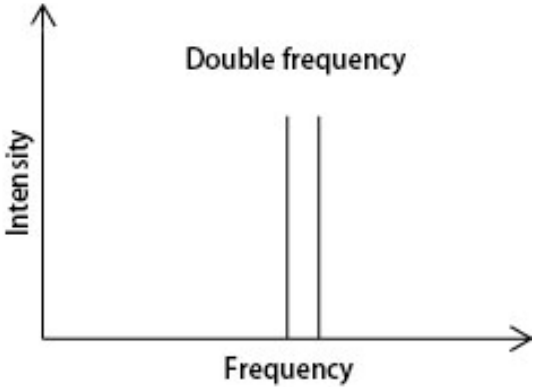
$$B(\sigma) = \int_0^\infty I(x)\cos(2\pi\sigma x)dx$$



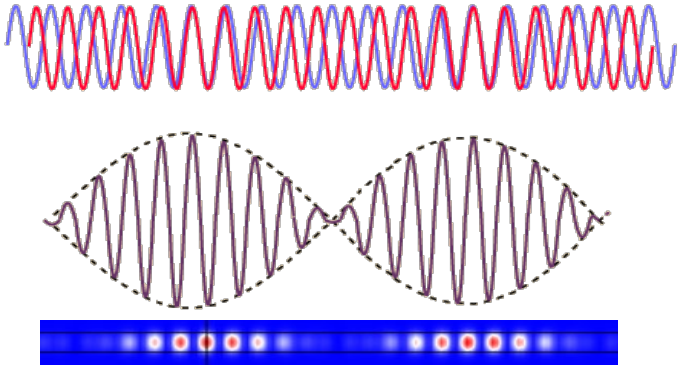
Example：Fourier Transform Spectroscopy



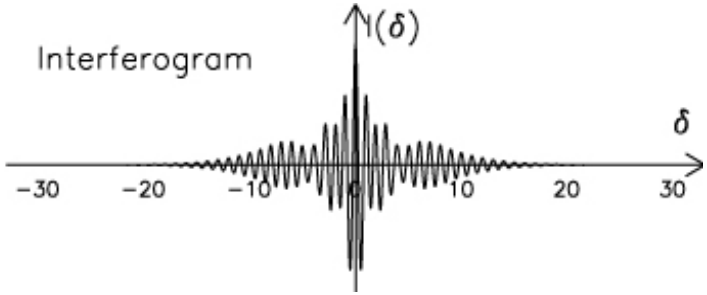




Beats

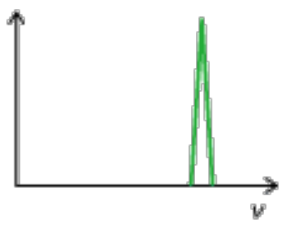


$$s(x) \sim A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{2\pi n x}{P}\right) + B_n \sin\left(\frac{2\pi n x}{P}\right)$$

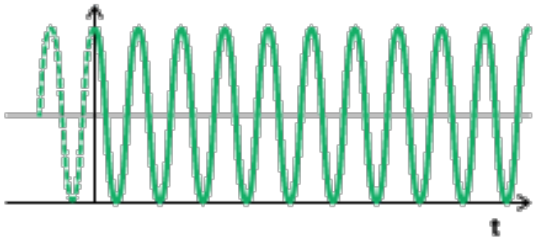


Example：Fourier Transform Spectroscopy

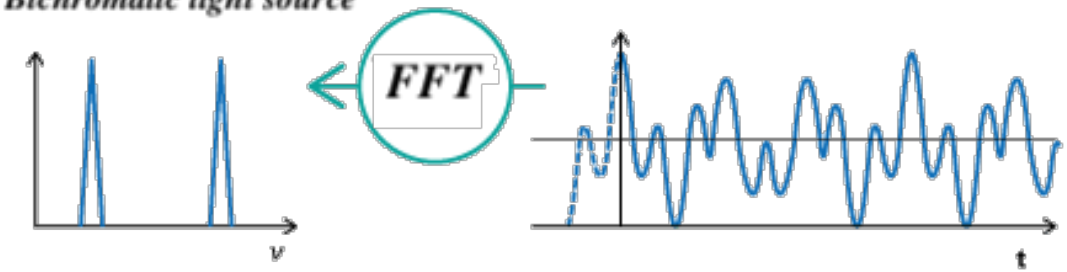
- *Monochromatic light source*



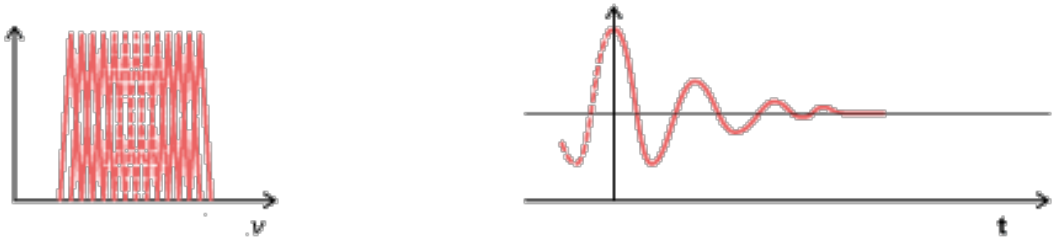
Interferogram



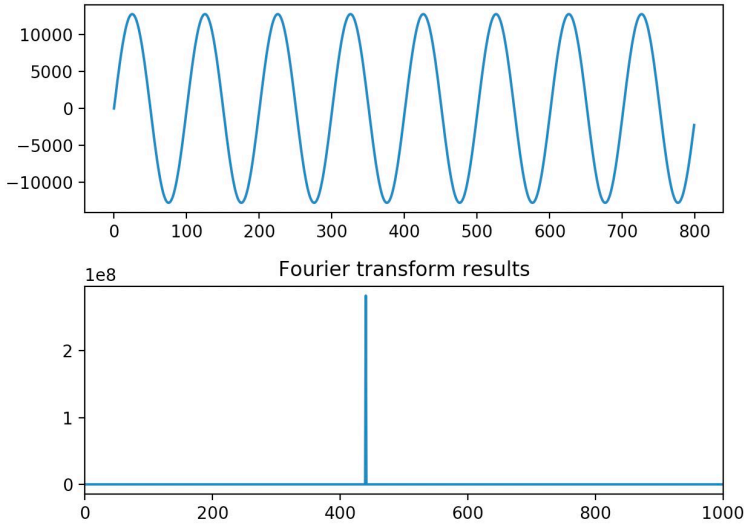
- *Bichromatic light source*



- *Continuous light source*



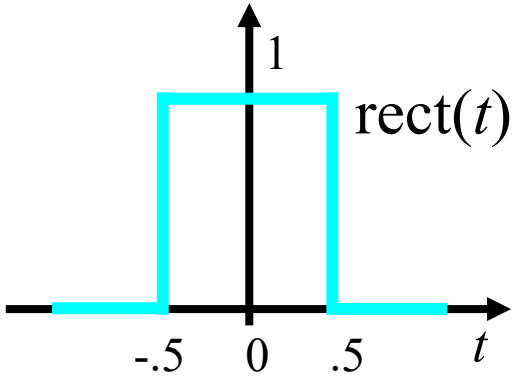
Example：Fourier Transform Spectroscopy



Infinite periodic function

⇩

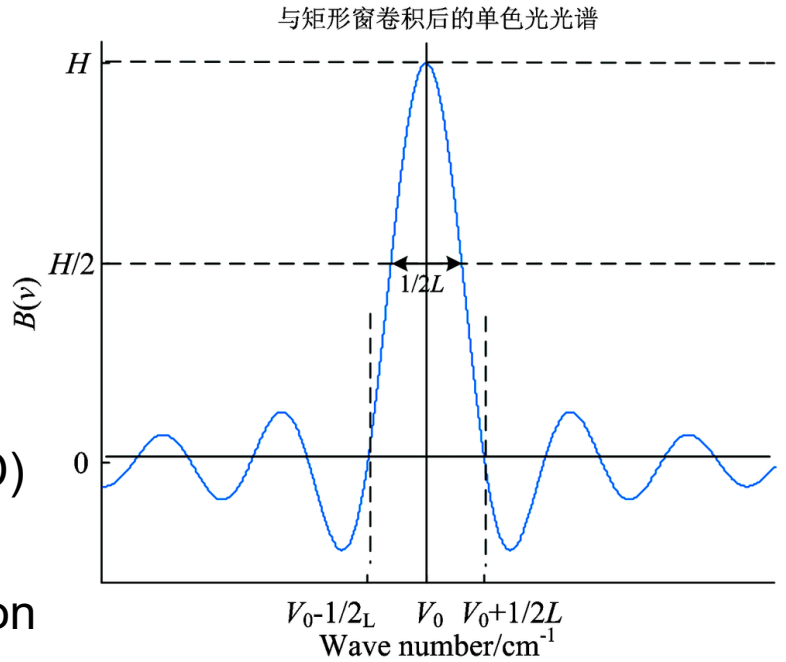
High spectral resolution



Short optical path difference (OPD)

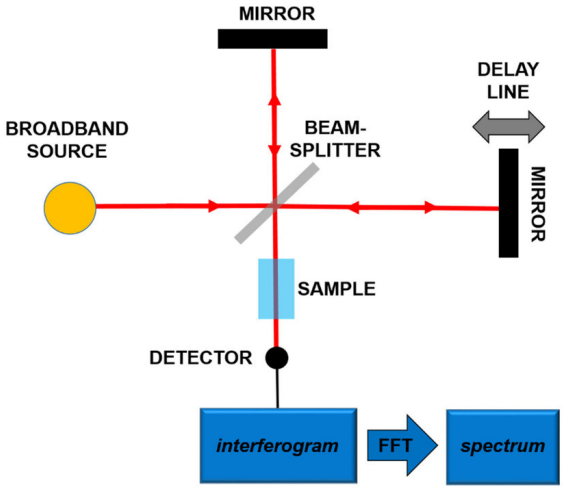
⇩

Convolution of rectangular function



Low spectral resolution

Example：Fourier Transform Spectroscopy



傅里叶变换光谱仪通常用于检测红外波段的吸收或发射光谱，工作原理是使用干涉仪来调制到达单个探测器上的光，再通过傅里叶变换将探测器上收集到的干涉图转换为与波长相关的光谱。傅里叶变换型光谱仪具有杂散光低、分辨率较高、光通量大、精度高等优点，使得微型化的同时满足高光通量和高分辨率等性能的要求。

思考：为什么红外光谱仪大部分是傅里叶型的，很少用光栅色散?

1. 多通道（Fellgett’s advantage）：所有光谱同时到达探测器，能量比色散/滤光后的强，减少探测时间，噪声较低。

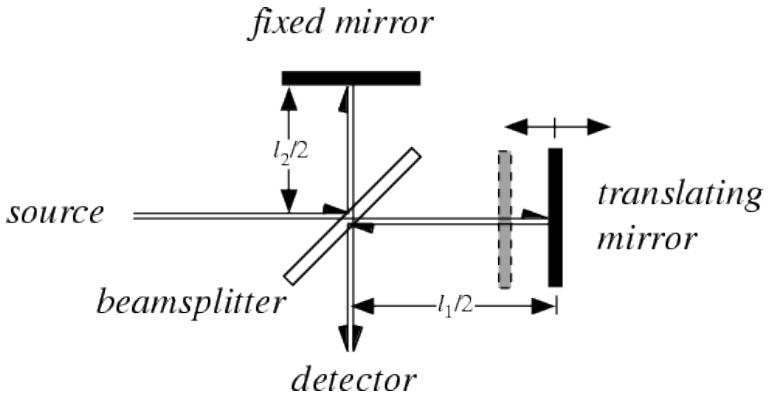
2. 高光通量（Jacquinot advantage）：光路简单，不需要狭缝，整体进光亮较高。

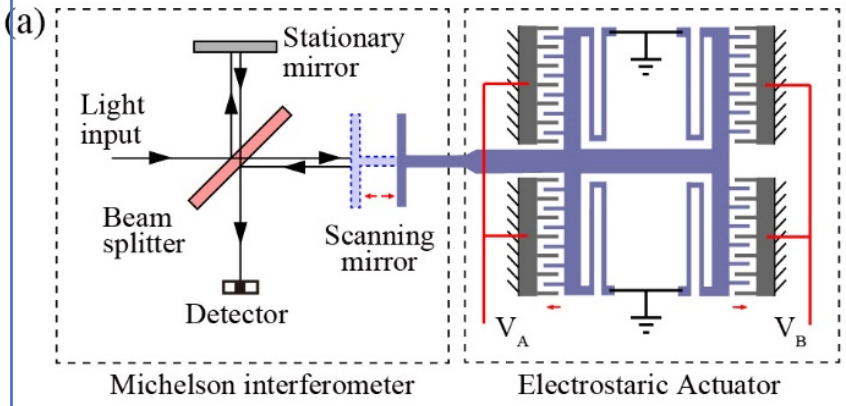
3. 高测量精度（Connes advantage）： FTIR光谱仪的光学结构简单,干涉仪只有一个动镜是运动部件,通常动镜是在无摩擦的空气轴承上移动,其运动又受高度稳定的He-Ne激光干涉系统监控,因此测量的重复性和准确度都很十分高。

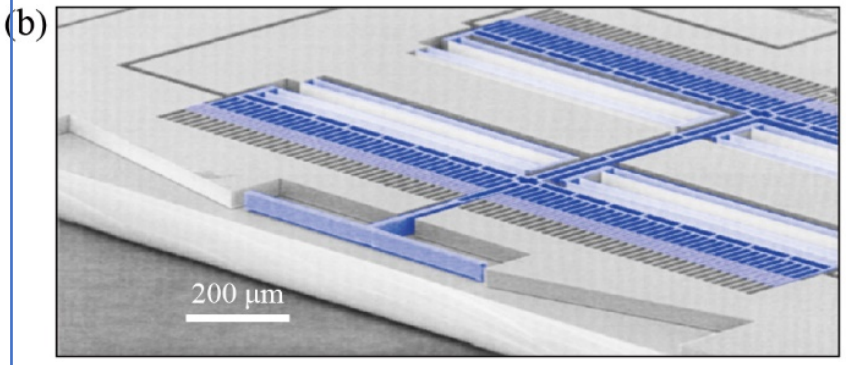
Example：Fourier Transform Spectroscopy

微型傅里叶光谱仪核心就是scan the Optical Path Difference (OPD),可分为两类：一类是**时间调制**型；另一类是**空间调制**型。







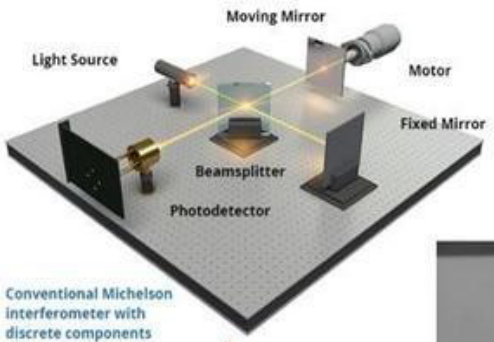


M. Herzig et al., ***Opt. Lett.*** 24, 1705 (1999)

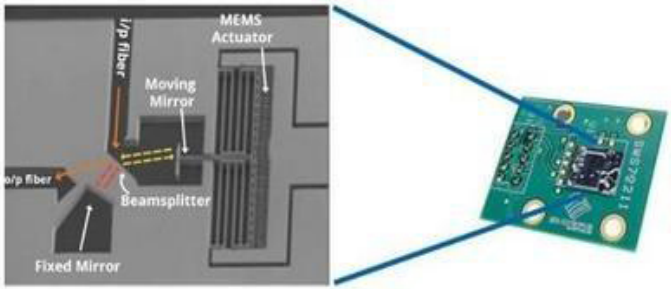
传统FTIR也是时间调制型，由驱动器控制迈克尔逊干涉仪的反射镜运动以获得干涉曲线。

通过MEMS工艺做成片上傅里叶型微光谱仪。动镜通过静电、电磁力、电热驱动移动。

Example：Fourier Transform Spectroscopy



Conventional Michelson interferometer with discrete components



Monolithic Michelson interferometer chip enable by SiMOST



美国Si-Ware公司的时间调制傅里叶型微光谱仪



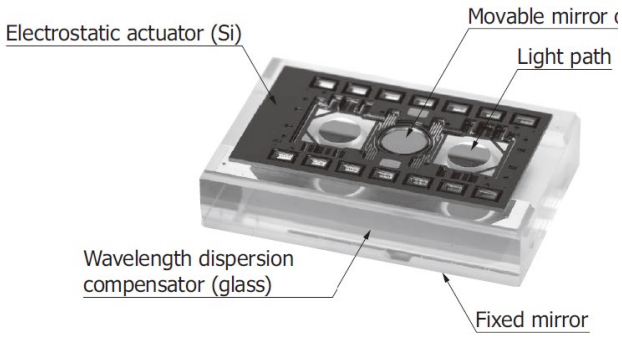
合肥南巢科技的时间调制傅里叶型微光谱仪， 6 nm分辨率

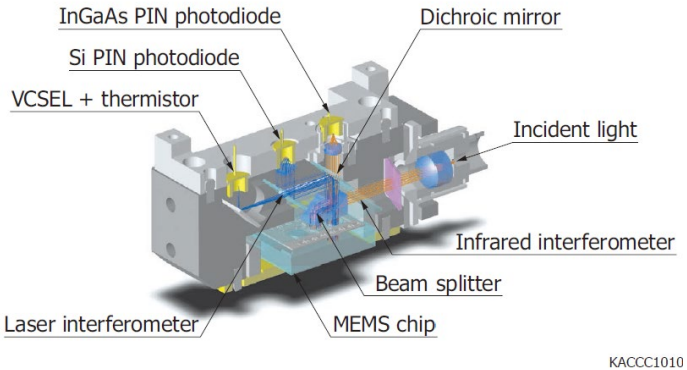
Example：Fourier Transform Spectroscopy

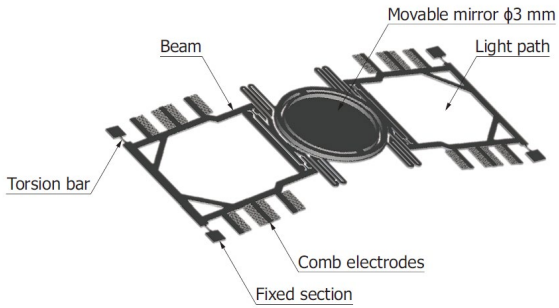


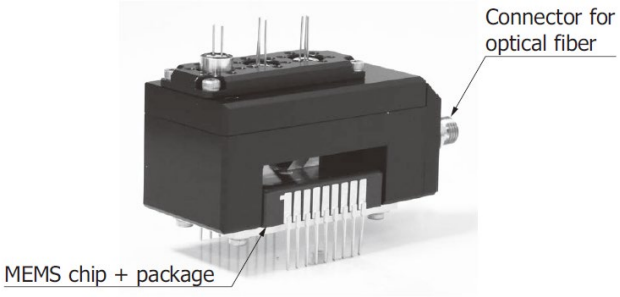
尺寸49×57×76 mm

分辨率5.7 nm





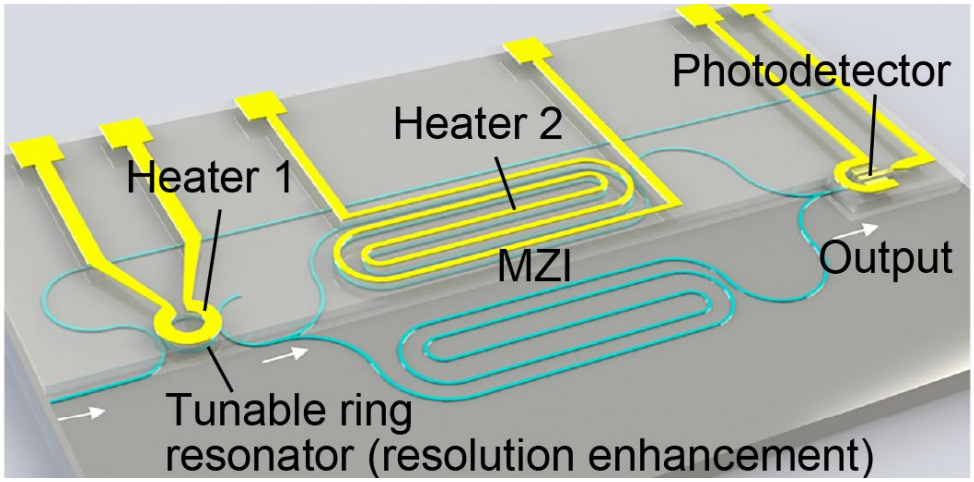




日本滨松公司的时间调制傅里叶型微光谱仪C15511-01

移动距离有限，分辨率有限，稳定性差

Example：Fourier Transform Spectroscopy



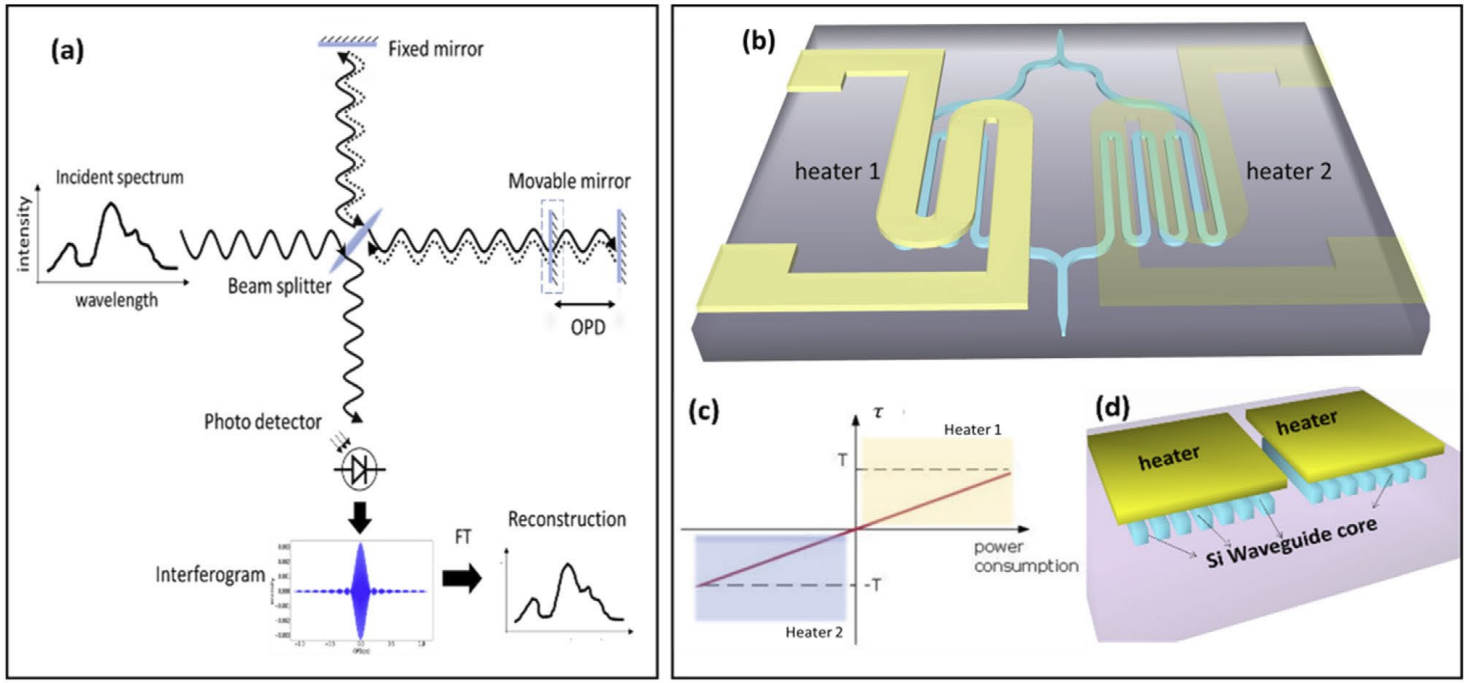
M. Souza *et al.*, ***Nature Communications*** 9, 665 (2018)

没有动镜的基于集成波导的平面片上傅里叶变换光谱仪。这些系统没有采用迈克尔逊结构，而是基于马赫-曾德尔干涉仪(MZIs)，Mach-Zehnder干涉仪中光被分裂到至少两个单向通道，然后进入同一个探测器并被重新组合，由于光在不同通道的光程差不同，因此相位差也不同，便产生了干涉。

连续调制的Mach-Zehnder干涉仪可以通过电光调制铌酸锂波导实现，也可以通过热光学效应实现，比如在集成光路附近嵌入微型加热器。

MZI 干涉仪，通过加热其中一个波导臂改变光程差，无移动部件，稳定

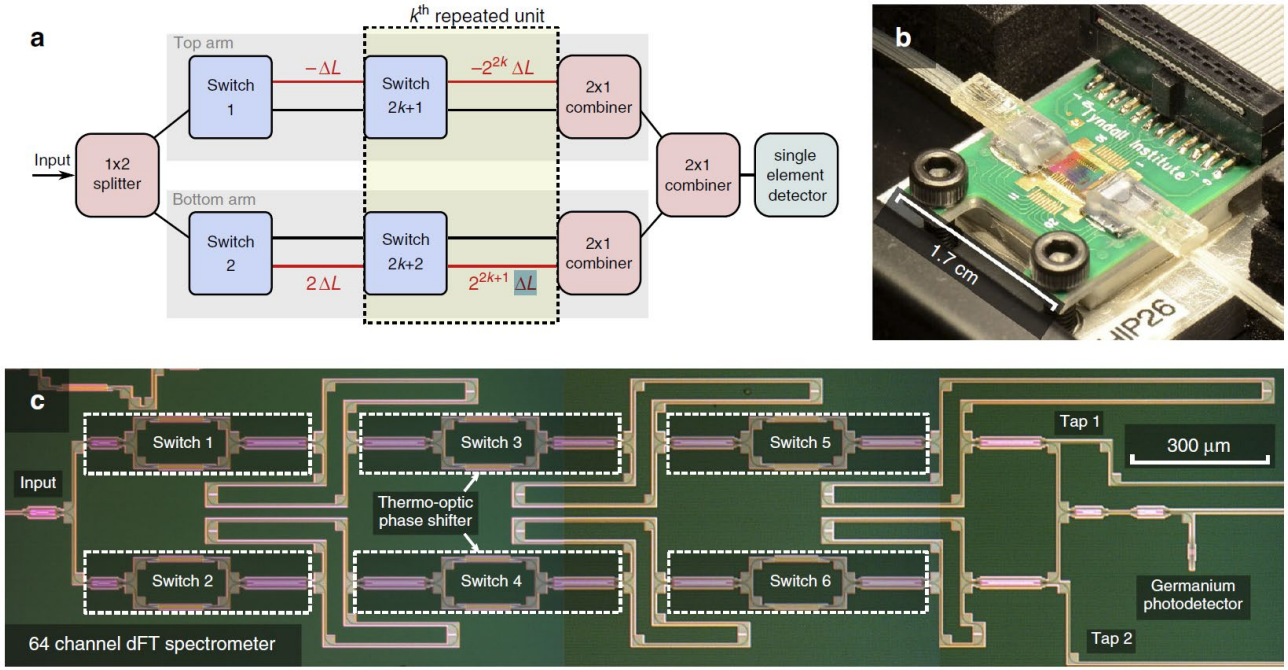
Example：Fourier Transform Spectroscopy



A. Li *et al*., ***Laser Photonics Review*** 15, 2000358 (2021)

光引科技的傅里叶型微光谱仪方案

Example：Fourier Transform Spectroscopy

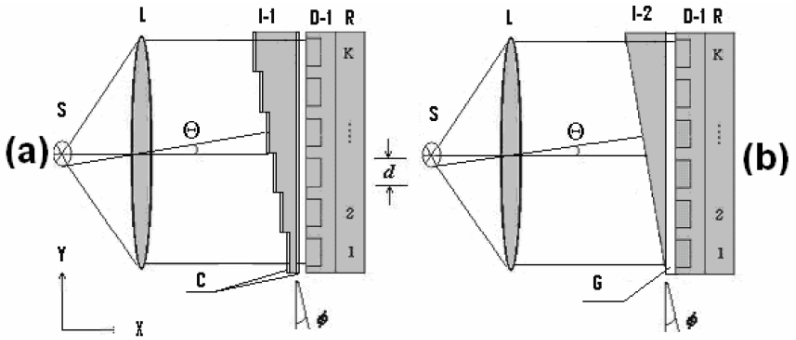


数字傅里叶变换光谱仪使用了一个具有许多光开光的光子电路，将光信号沿着不同长度的路径传递。在此，光谱分辨率取决于路径的数量或光谱通道数，其与光开关的数量成指数关系。

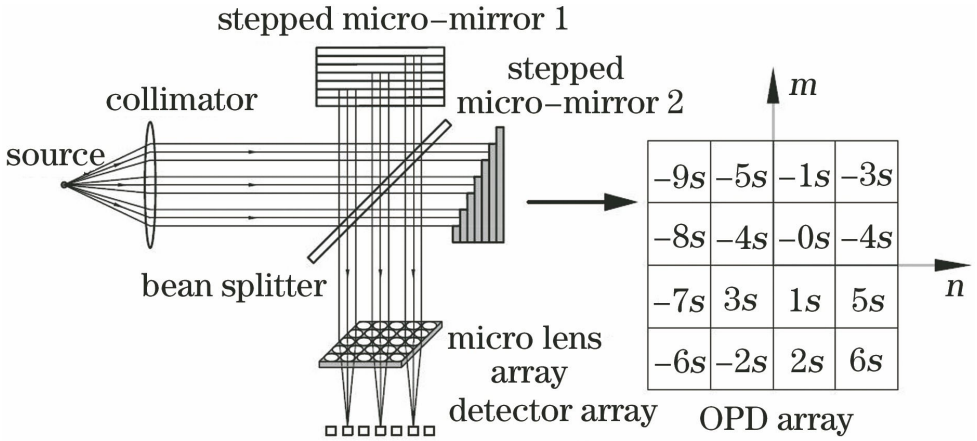
D. Kita *et al.*, ***Nature Communications*** 9, 4405 (2018)

MIT的数字MZI 干涉仪，通过多个光开关调节光路从而调节光程差，无移动部件

Example：Fourier Transform Spectroscopy



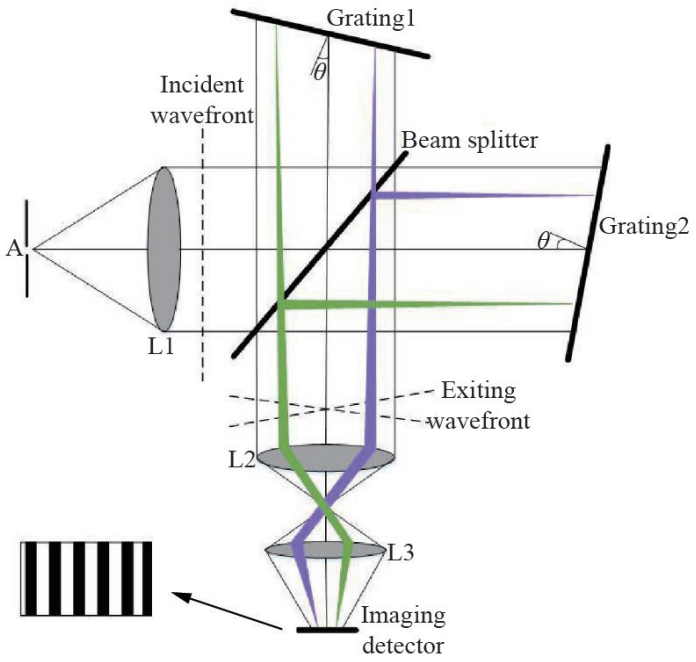
A. Manuilskiy *et al*., ***SPIE*** 6395, 639504 (2006)



吕金光 等. ***光学学报***, 38, 0230001 (2018)

静态傅里叶型光谱仪，通过空间位置的改变得到光程差序列，无须扫描实现干涉信号的空间调制。

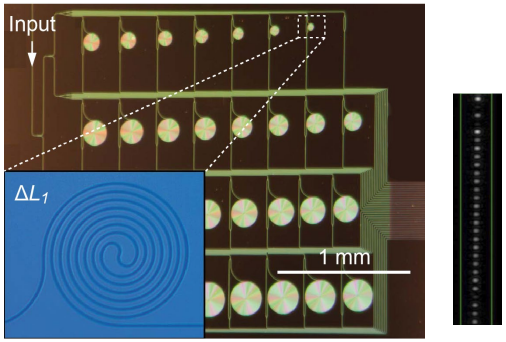
Example：Fourier Transform Spectroscopy



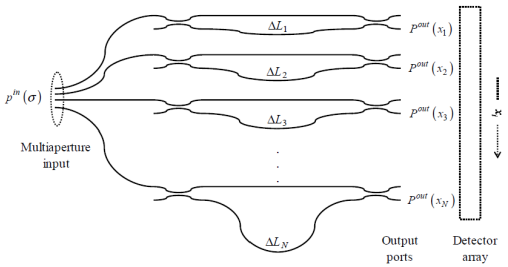
王新强 等. ***应用光学,*** 41, 462 (2020)

空间外差型，通过两个光栅对不同波长的路径差，实现干涉信号的空间调制。

Example：Fourier Transform Spectroscopy



A. Velasco *et al.*, ***Optics Letters*** 38, 706 (2013)



M. Florjanczyk *et al.*, ***Optics Express*** 26, 18178 (2007)

不同光程差的多个Mach-Zehnder干涉仪阵列空间调制。

基于多个Mach-Zehnder干涉仪阵列，形成空间调制。例如，螺旋状的波导阵列，波导的长度线性变化，从而引起光程差。

但是，此类系统的光谱分辨率受到最大光程差以及Mach-Zehnder干涉仪数量的限制，这都限制了光谱仪在保持高性能的同时尽可能的微型化。

Table 1. ANN Classification: Best Results Confusion Matrix

	Predicted			
Actual	Class 1	Class 2	Class 3	Class 4
Class 1	1782	383	161	1
Class 2	90	1869	318	49
Class 3	0	676	1647	1
Class 4	1	39	62	2222

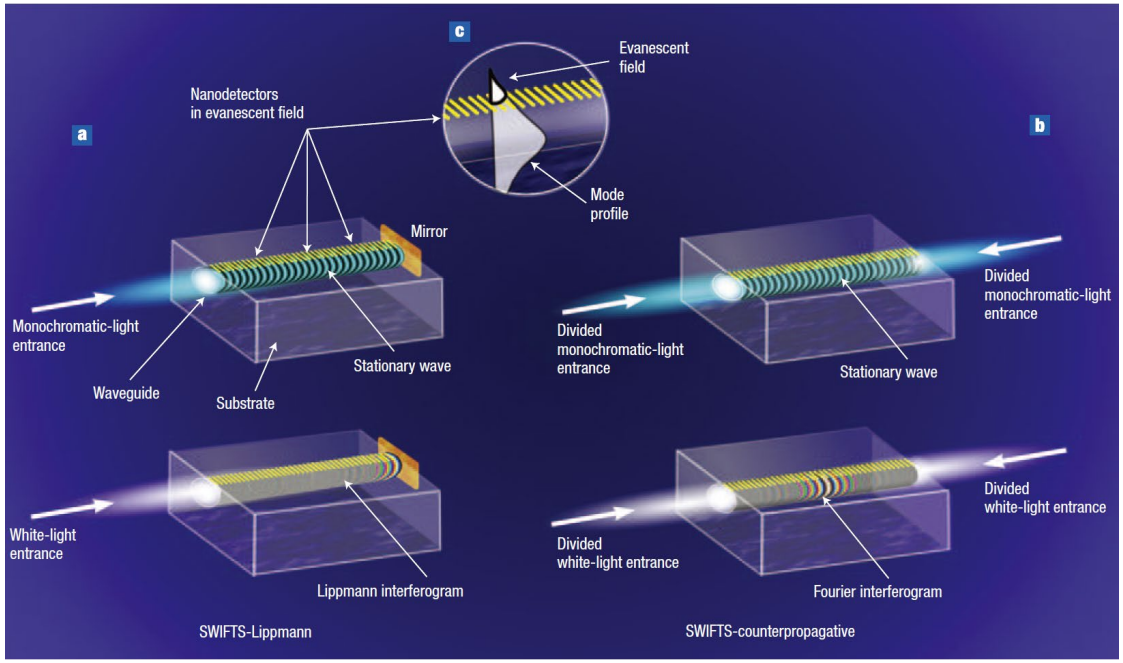
Table 2. SVM Classification: Best Results Confusion Matrix

	Predicted			
Actual	Class 1	Class 2	Class 3	Class 4
Class 1	1523	343	461	0
Class 2	361	1622	343	0
Class 3	133	5	2186	0
Class 4	0	0	0	2324

A. Bermello *et al.*, ***Optics Letters*** 44, 5840 (2019)

机器学习提高光谱范围和分辨率。

Example：Fourier Transform Spectroscopy



E. Coarer *et al*., ***Nature Photonics*** 1, 473 (2007)

驻波集成傅里叶变换光谱仪

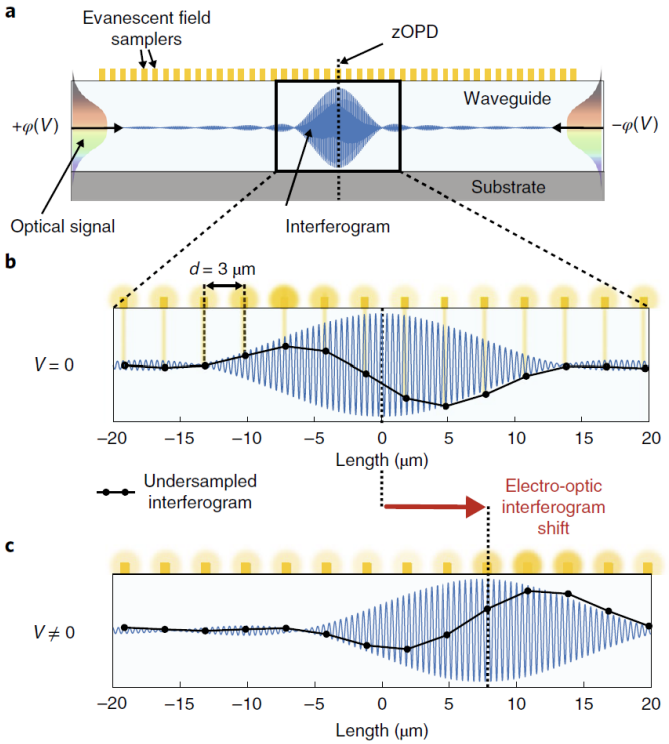
驻波集成傅里叶变换光谱仪有两种工作方式：

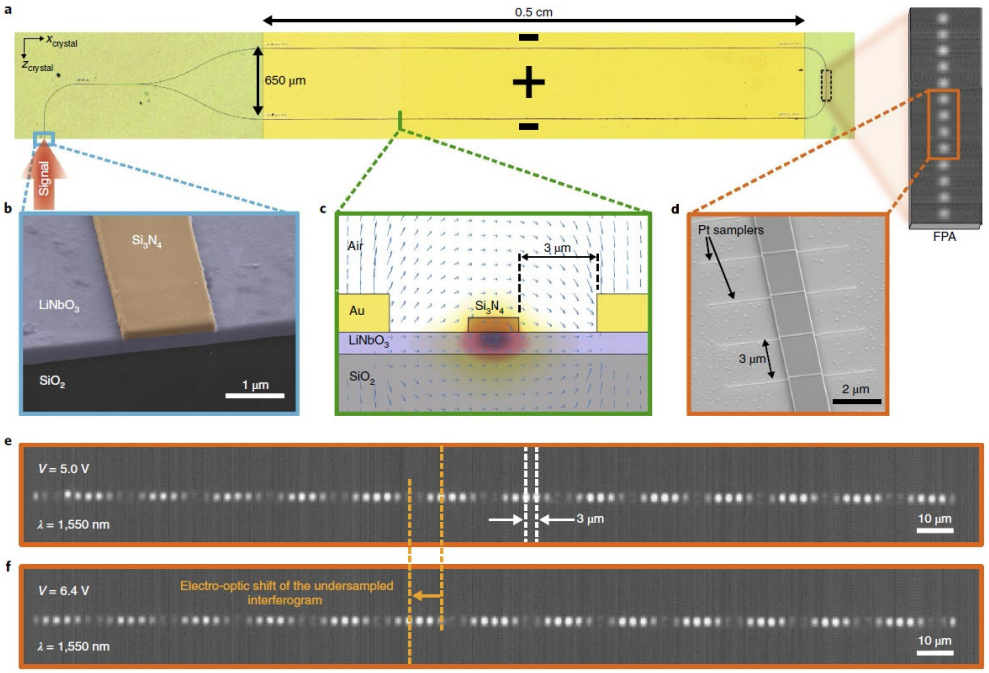
一种为单反式，即进入波导的光经反射镜反射后与入射光干涉，形成驻波；

另一种为相遇式，即从波导的两边分别入射光波，干涉形成驻波。将微型探测器集成到波导上，每个像元就可以采集到一定的干涉信号。

探测器像元尺寸限制了系统的光谱分辨率。

Example：Fourier Transform Spectroscopy



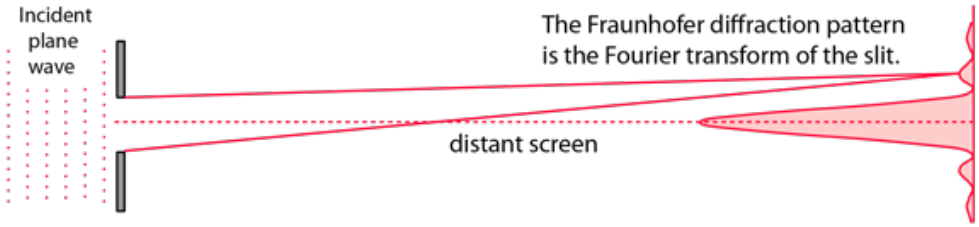


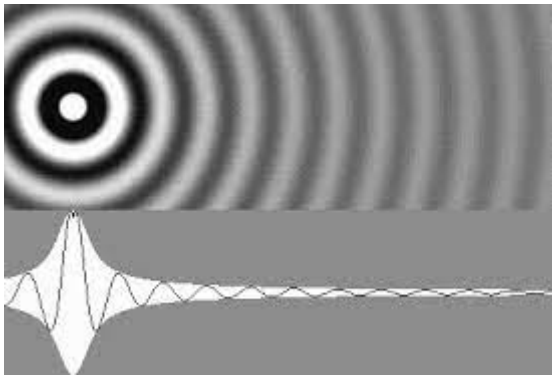
D. Pohl *et al.*, ***Nature Photonics*** 14, 24 (2020)

干涉位置通过电压可调，解决驻波集成傅里叶变换光谱仪欠采样问题，提高分辨率

Example 2: Fourier Optics

Fourier optics methods can be visualized by considering the Fraunhofer diffraction pattern of a single slit. The diffraction process transforms the slit in the object plane to a diffraction pattern in the distant image plane. This diffraction pattern contains information about the slit in a form in which smaller spatial details (narrower slits) are transformed into larger spatial displacement in the image plane (broader diffraction patterns). Smaller spatial detail can be referred to as a higher "spatial frequency", and the diffraction pattern produces a plot in which greater distance from the optic axis implies greater spatial frequency. This kind of transformation, where a plot of light distribution is transformed into plot of spatial frequency is an example of a Fourier transformation and is a conceptual starting point for Fourier optics.





Fourier transform of the single slit (rectangular function)

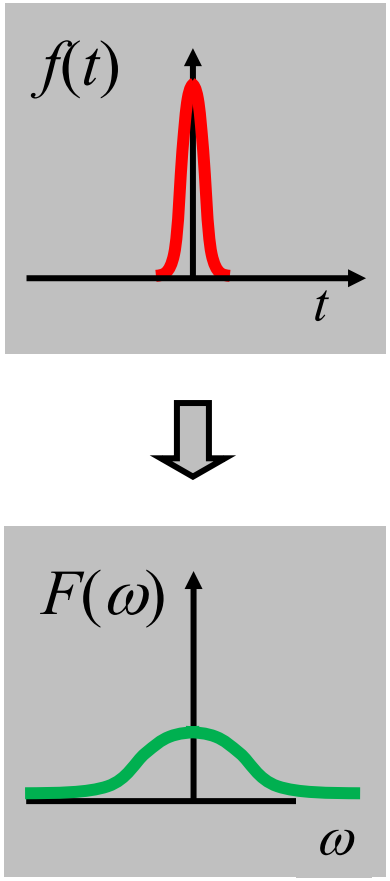
Fourier Transform with Respect to Space

If $f(x)$ is a function of position x, we can define a **spatial** Fourier transform from x to k.

We refer to k as the **spatial frequency**.

$$F(k) \equiv \int_{-\infty}^{\infty} f(x) \exp(-ikx)\, dx$$

$$\mathrm{F}\ \{f(x)\} = F(k)$$



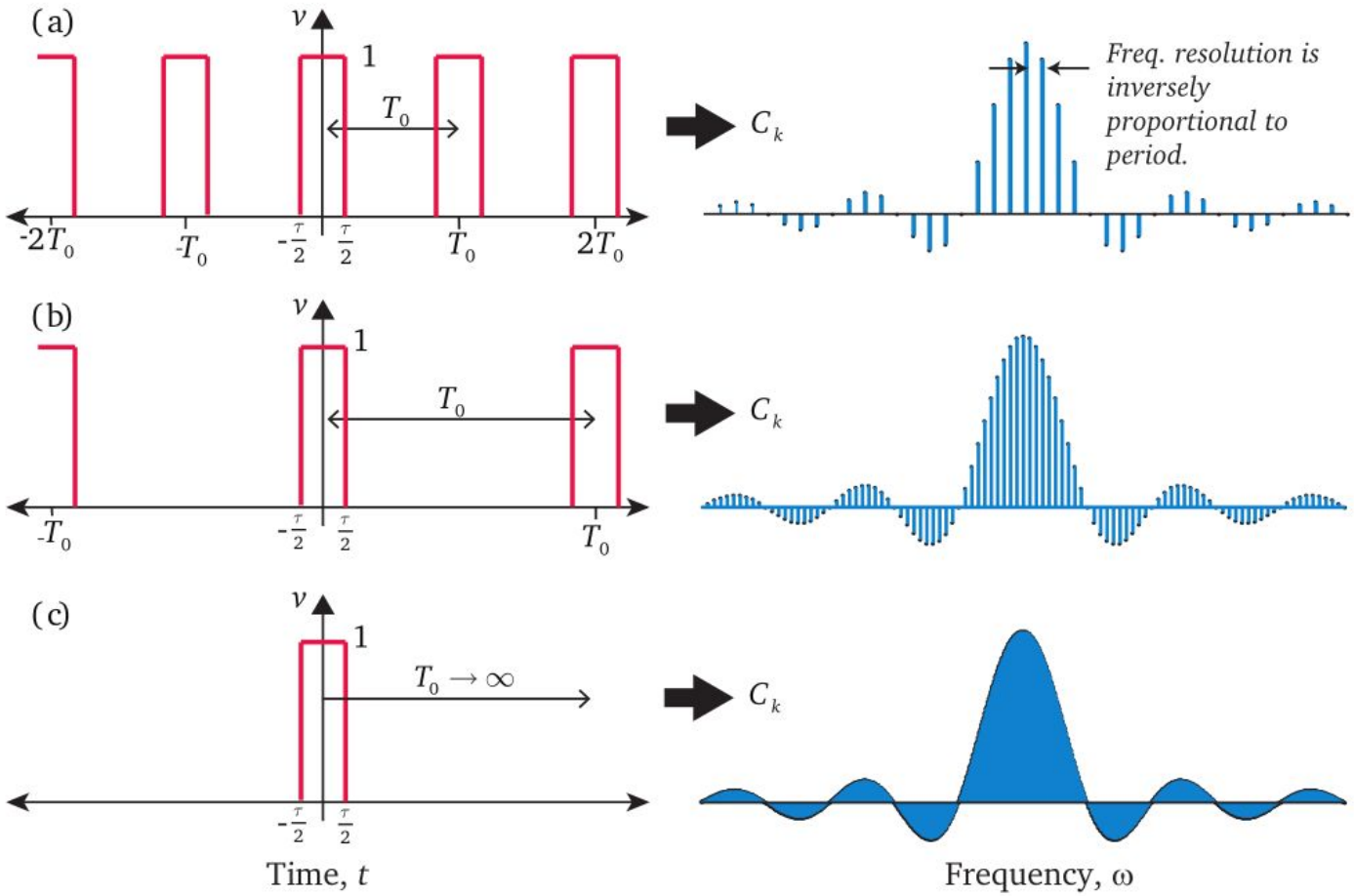
And the **inverse** spatial Fourier transform is now:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(k) \exp(ikx)\, dk$$

Everything we've said about Fourier transforms between the t and ω domains also applies to the x and k domains.

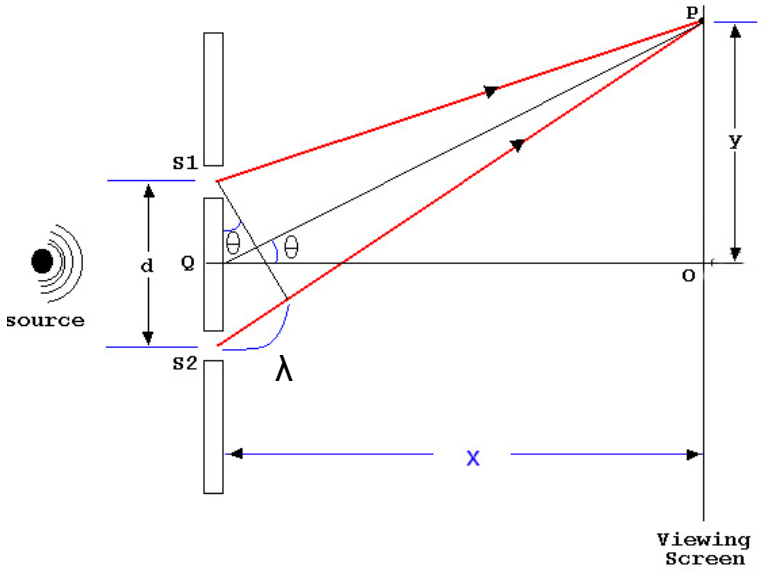
Fourier series and Fourier transform

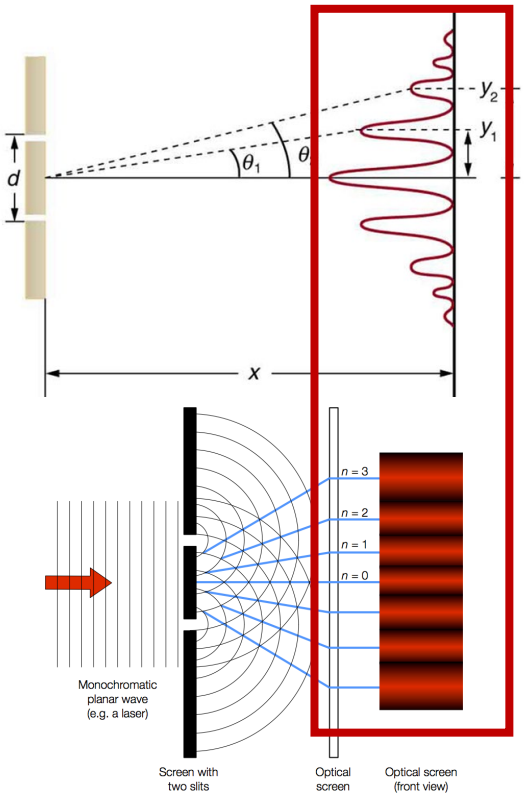
Fourier series for periodic functions, and Fourier transform for functions that are not periodic. The Fourier transform is a generalization of Fourier series.

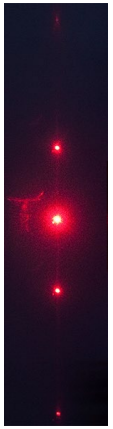


Example：Fourier Optics

Double-slit interference







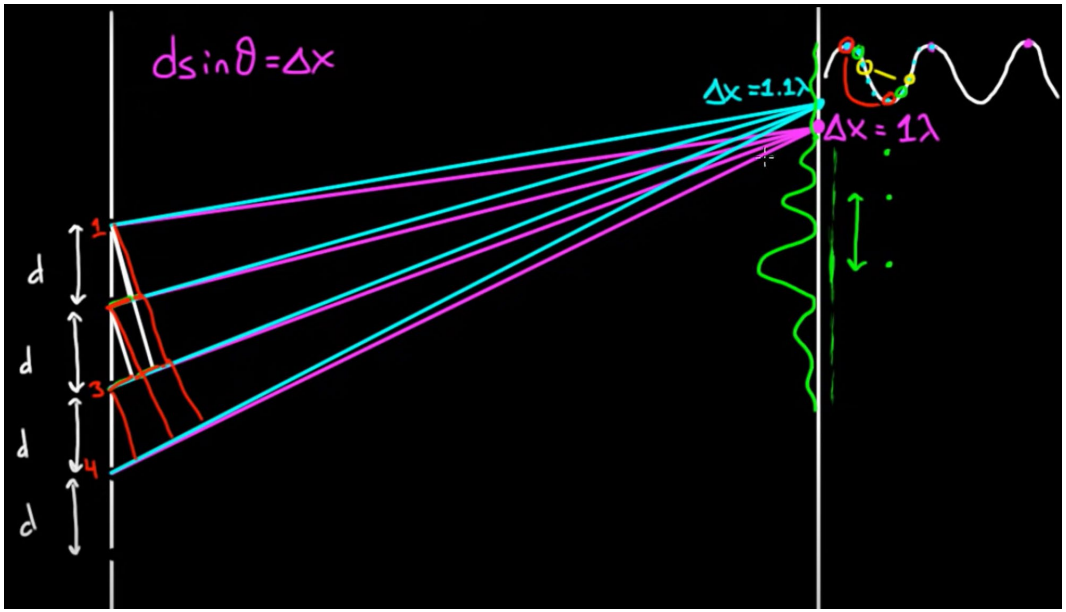
双缝干涉条纹交叠，而光栅的干涉图案是分立的点或线

With the assumption that x >> d, we can write the condition for constructive interference as:

$$m\lambda \approx d \sin\theta$$

where m = 0 (θ = 0), 1 (θ_1) , 2 (θ_2) ,.....

Example：Fourier Optics



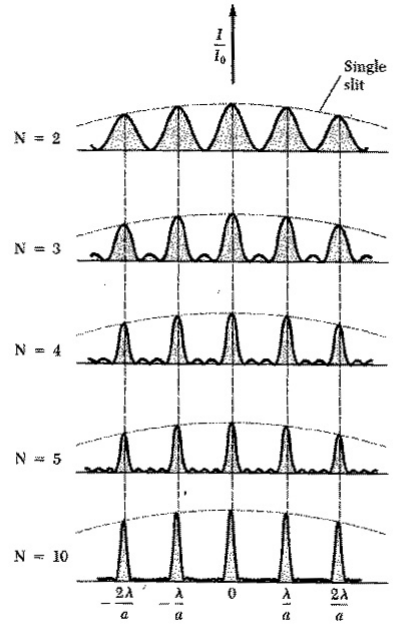
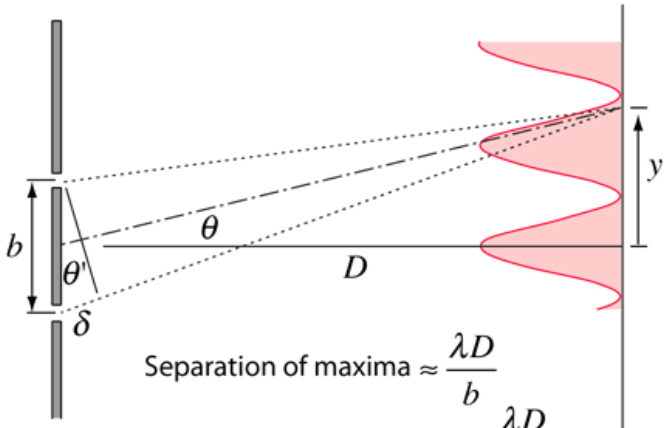


Figure 37.13 Multiple-slit interference patterns. Note that as the number of slits is increased, the primary maxima become narrower but remain fixed in position. Furthermore, the number of secondary maxima increases as the number of slits is increased. The decrease in intensity of the maxima, as indicated by the dashed lines, is due to the phenomenon of diffraction, which is discussed in Chapter 38.

Example：Fourier Optics

$$m\lambda = b\sin\theta \approx b\frac{y}{D} \quad \text{so} \quad \Delta y \approx \frac{Dm\Delta\lambda}{b}$$



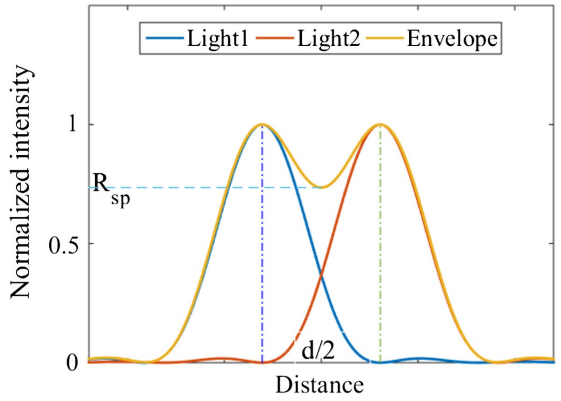
Separation of maxima $\approx \frac{\lambda D}{b}$

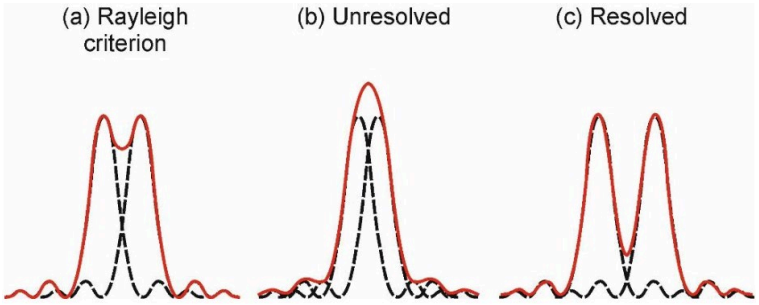
Distance to first minimum $\Delta y \approx \frac{\lambda D}{Nb}$

Substituting $\frac{\lambda D}{Nb} \approx \frac{Dm\Delta\lambda}{b}$ gives

Resolvance $\boxed{R = \frac{\lambda}{\Delta\lambda} = mN}$

where N(每毫米刻线数) is the total number of slits illuminated and m is the order of the diffraction.



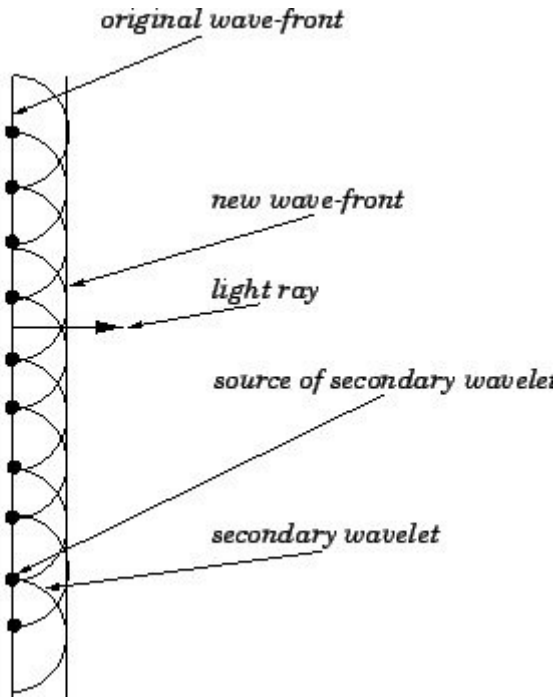


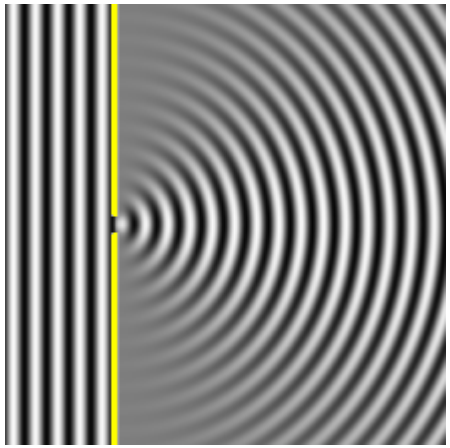
光谱分辨率判断标准-Rayleigh Criterion：

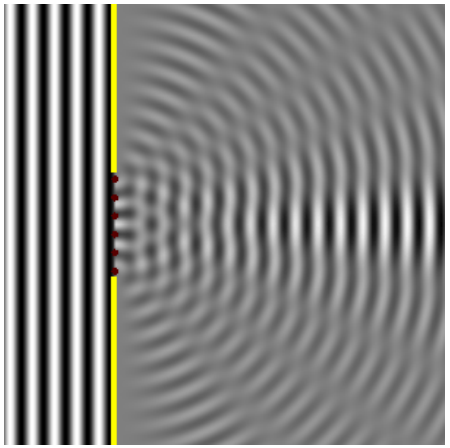
两个峰间距等于其中一个峰的半高全宽

Example：Fourier Optics 数学描述

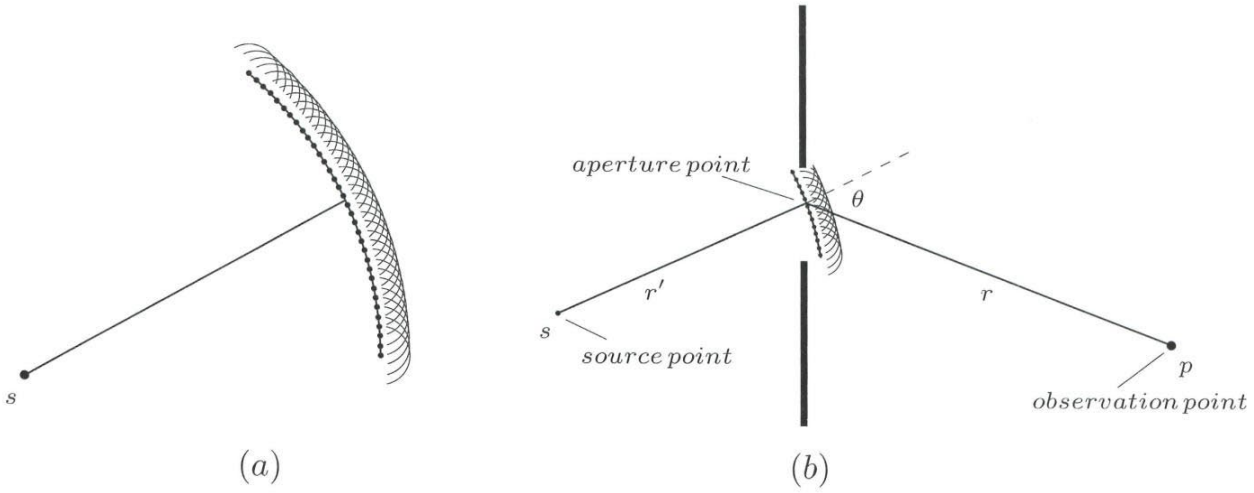
The Huygens–Fresnel principle (named after Dutch physicist Christiaan Huygens and French physicist Augustin-Jean Fresnel) states that every point on a wavefront is itself the source of spherical wavelets, and the secondary wavelets emanating from different points mutually interfere. The sum of these spherical wavelets forms a new wavefront.





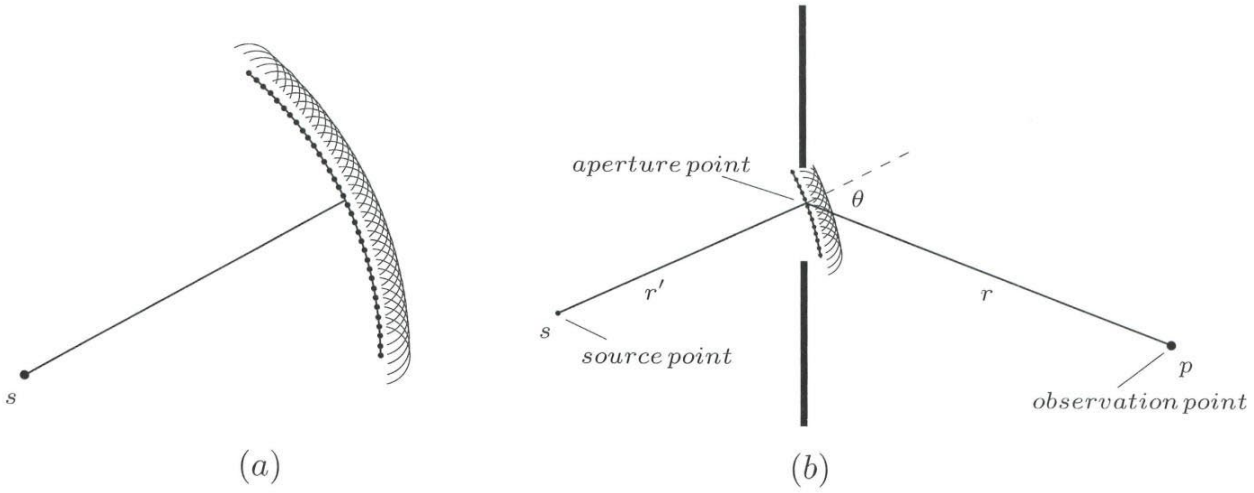


Example：Fourier Optics 数学描述



- s点在小孔处波前电场可描述为：$E_A(r',t)=\frac{E_0}{r'}e^{i(kr'-\omega t)}$，$t$=0时为 $E_A=\frac{E_0}{r'}e^{i(kr')}$；
- 小孔上的点作为次波源在p处的球面波电场为 $dE_p=\frac{E_A}{r}e^{i(kr-\omega t)}=\frac{E_0}{r'r}e^{ik(r+r')}e^{-i\omega t}$；
- 所有点的合电场为 $E_p=E_0e^{-i\omega t}\int_A\frac{e^{ik(r+r')}}{r'r}dA$；
- 更为精确的数学描述 (Fresnel-Kirchhoff)：$E_p=i\frac{A}{\lambda}e^{-i\omega t}\int_A\left[\frac{1+\cos\theta}{2}\right]\frac{e^{ik(r+r')}}{r'r}dA$。

Example：Fourier Optics 数学描述

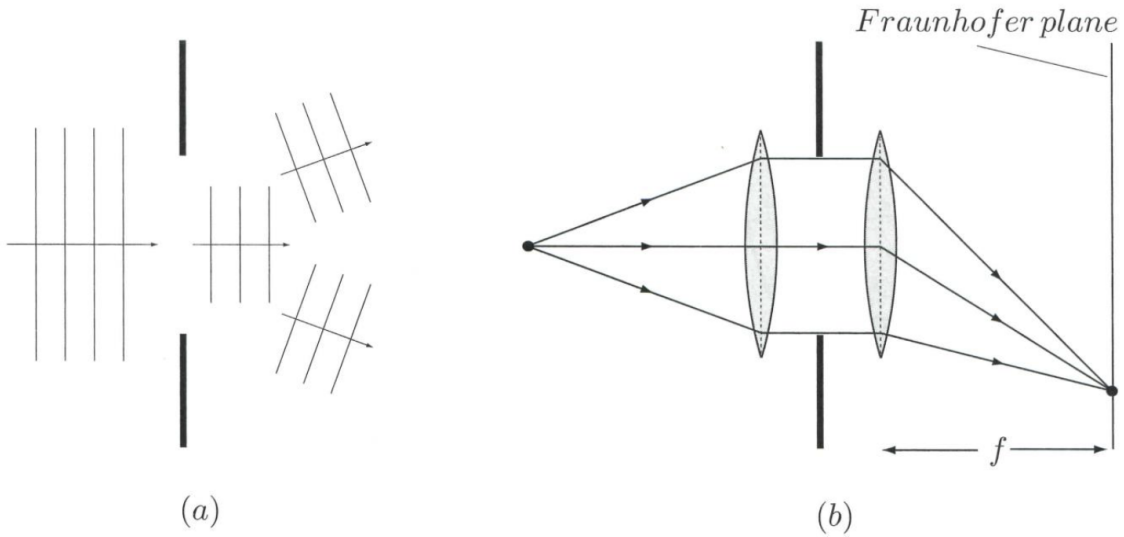


$E_p = i\frac{A}{\lambda}e^{-i\omega t}\int_A \left[\frac{1+\cos\theta}{2}\right]\frac{e^{ik(r+r')}}{r'r}dA$，其中 $\frac{1+\cos\theta}{2}$ 为倾斜因子 (obliquity factor)。

- 只有在简单假设情况下，上式才可以求解。
- 如果对于s点和p点角度θ较小，倾斜因子可近似为1。
- 如果s点和p点在较远距离，入射和衍射场可看作平面波，为远场或**Fraunhofer衍射**。
- 如果s点或/和p点距离小孔较近，入射或/和衍射场曲率不可忽略，为近场或**Fresnel衍射**。

Example：Fourier Optics

Fraunhofer衍射：入射波和衍射波波前曲率可以忽略，或者说入射光源和衍射光观察点距离小孔距离足够远。



- Faunhofer衍射采用上页公式 $E_p = i\frac{A}{\lambda}e^{-i\omega t}\int_A \left[\frac{1+\cos\theta}{2}\right]\frac{e^{ik(r+r')}}{r'r}dA$ 并假设：

1、衍射波角分布小，obliquity factor $\frac{1+\cos\theta}{2}$ 可被忽略；

2、$\frac{e^{ikr'}}{r'}$ 项在小孔处为常量，因此可以从积分中提出；

3、$\frac{e^{ikr}}{r}$ 的变化主要来自于指数项，r可定为常量。

$$E_p = C\int_A e^{ikr}dA$$

Fraunhofer衍射积分

Example：Fourier Optics

双狭缝衍射：width *b*, length *L*, separation *d*; *dA=L dy*

- 此时有：$E_p = C\int_0^b e^{ik(r_0+y\sin\theta)}Ldy + C\int_d^{d+b} e^{ik(r_0+y\sin\theta)}Ldy = CLe^{ikr_0}\left[\left(\frac{e^{iky\sin\theta}}{ik\sin\theta}\Big|_0^b\right)+\left(\frac{e^{iky\sin\theta}}{ik\sin\theta}\Big|_d^{d+b}\right)\right]$

$$= \frac{CLe^{ikr_0}}{ik\sin\theta}\left[e^{ikb\sin\theta} - 1 + e^{ik(d+b)\sin\theta} - e^{ikd\sin\theta}\right] = \frac{CLe^{ikr_0}}{ik\sin\theta}\left(e^{ikb\sin\theta}-1\right)\left(1+e^{ikd\sin\theta}\right)$$

$$= \frac{CLe^{ikr_0}b}{\frac{1}{2}kb\sin\theta}(2)\,e^{i\frac{1}{2}kb\sin\theta}e^{i\frac{1}{2}kd\sin\theta}\left(\frac{e^{i\frac{1}{2}kb\sin\theta}-e^{-i\frac{1}{2}kb\sin\theta}}{2i}\right)\left(\frac{e^{i\frac{1}{2}kd\sin\theta}+e^{-i\frac{1}{2}kd\sin\theta}}{2}\right) = C'\left(\frac{\sin\beta}{\beta}\right)(\cos\gamma)$$

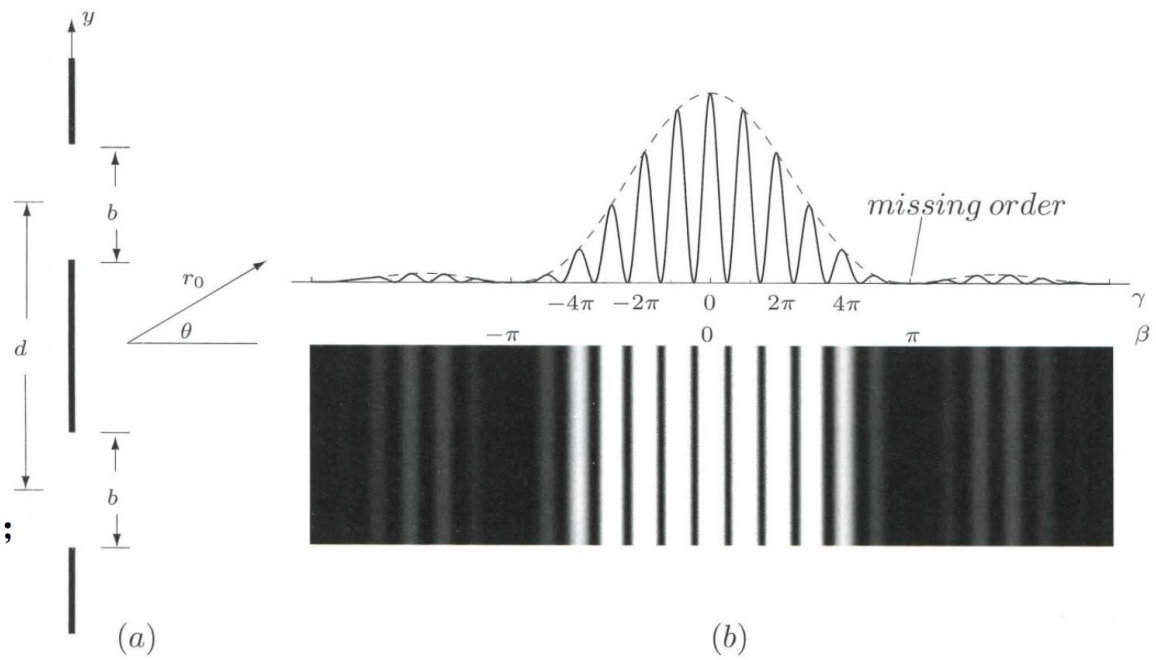
- 其中 $\beta = \frac{1}{2}kb\sin\theta$，$\gamma = \frac{1}{2}kd\sin\theta$;

- 衍射图样的辐射度为：

$$I = I_0\left(\frac{\sin\beta}{\beta}\right)^2(\cos\gamma)^2 \;;$$

- Young's双缝假设无限窄狭缝，衍射图样可以理解为两个狭缝各自衍射场的干涉；

- 亮条纹在 $\gamma = m\pi$ 处，*m*为整数；envelop为零处 $\beta = m\pi$，missing order。



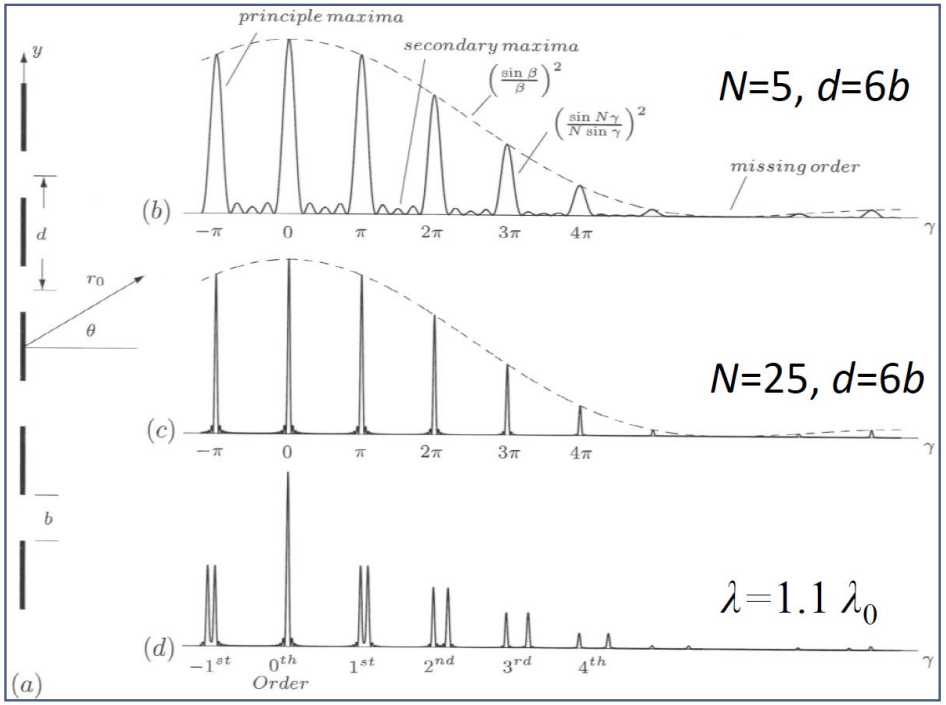
Example：Fourier Optics

N狭缝衍射： width *b*, length *L*, separation *d*; *dA* = *Ldy*

衍射场：
$$E_p = Ce^{ikr_0}L\left[\int_0^b e^{iky\sin\theta}dy + \int_d^{d+b} + \int_{2d}^{2d+b} + \ldots + \int_{(N-1)d}^{(N-1)d+b} e^{iky\sin\theta}dy\right] = Ce^{ikr_0}L\frac{\left(e^{ikb\sin\theta}-1\right)}{ik\sin\theta}\left[\frac{1-e^{ikNd\sin\theta}}{1-e^{ikd\sin\theta}}\right]$$

$$= Ce^{ikr_0}L2be^{i\beta}e^{i(N-1)\gamma}\frac{\sin\beta}{\beta}\left[\frac{\sin N\gamma}{\sin\gamma}\right]$$

$$= E_0\frac{\sin\beta}{\beta}\left[\frac{\sin N\gamma}{N\sin\gamma}\right];$$



- 其中 $\beta = \frac{1}{2}kb\sin\theta \qquad \gamma = \frac{1}{2}kd\sin\theta$ ；
- 辐照度 $I = I_0\left(\frac{\sin\beta}{\beta}\right)^2\left(\frac{\sin N\gamma}{N\sin\gamma}\right)^2$；
- Envelop $\left(\frac{\sin\beta}{\beta}\right)^2$；
- 主极大：$\gamma = \frac{1}{2}kd\sin\theta = m\pi$，或 $d\sin\theta = m\lambda$ ；

$$\lim_{\gamma\to m\pi}\frac{\sin N\gamma}{\sin\gamma} = \lim_{\gamma\to m\pi}\frac{N\cos N\gamma}{\cos\gamma} = \pm N$$

Example：Fourier Optics

衍射光栅：特殊设计的小孔，具有明显的衍射主极大，用于光谱测量中。

- 垂直入射时有：$d\sin\theta = m\lambda$；

  角度入射有：$d\left(\sin\theta_i + \sin\theta\right) = m\lambda$；

  光栅方程

- 同一级别，不同波长具有不同的衍射角度，位于衍射面的不同空间位置；

